BPUT Hackathon 2026 — PS05 Final Technical Report

GPS-Denied 3D Autonomous UAV with LiDAR–Camera Fusion, 3D Planning and Precision Landing

Generated by `scripts/generate_report.py` from logged simulation data (repository revision `75d4b5d`). Every number in sections 28–34 is read from `results/`; anything not measured is marked NOT MEASURED.

2. Abstract

A simulated 5–7 kg-class quadrotor (PX4 X500 airframe) flies a 50 m × 50 m 3D obstacle course without GPS. Position comes from the PS05 external tracking station (r, θ, φ) fused by PX4 EKF2; a forward RGB camera and a 360° 3D LiDAR are time-synchronised and fused into an RGB point cloud that builds a ray-traced 3D occupancy map. A weighted 3D A* plans inside an approved mission corridor around inflated obstacles; a local layer stops and replans when new obstacles appear on the path. A downward camera detects an AprilTag and a downward flash LiDAR validates the landing surface before and during a staged precision descent. Across 61 logged simulation trials the horizontal landing error over all landed runs was mean 4.4 cm, RMSE 5.0 cm, P95 8.2 cm (n = 57), with 0 collisions.

3. Problem statement

PS05 requires GPS-independent autonomous guidance of a UAV in a 3D environment: determine the vehicle state without satellite positioning, keep it inside a permissible corridor around an approved trajectory ($E = \sqrt{\Delta X^2+\Delta Y^2+\Delta Z^2} \le E_{max}$, otherwise critical alarm and containment), and — per the implementation guides — perceive obstacles with LiDAR and camera, plan and replan in 3D, and land precisely on a marked pad after validating it.

4. Motivation

GNSS is unreliable in urban canyons, under canopy and under jamming/spoofing. A 5–7 kg UAV failing over people is hazardous, so the system must be predictable (approved corridor), able to react to the unexpected (perception + replanning) and conservative at touchdown (independent geometric validation).

5. Requirements

Derived from the three supplied PS05 documents; the full requirement-by-requirement matrix with evidence is `docs/REQUIREMENTS_TRACEABILITY.md`. Key quantitative targets: no GPS in the guidance chain; tracking latency < 20 ms; telemetry > 50 Hz; mean simulated trajectory error E < 0.5 m; landing error, success rate, clearance, latencies and collisions reported over 20–30+ trials.

6. System architecture

Gazebo Harmonic simulates world, physics and sensors; PX4 v1.17.0 SITL is the flight controller; ROS 2 Jazzy carries all perception, mapping, planning and landing logic; the web dashboard is a read-only observer. Data flow and package map: `docs/ARCHITECTURE.md`, ROS graph: `docs/ROS_GRAPH.md`.

7. PX4 architecture

PX4 runs in lockstep with Gazebo (standalone gz mode). Custom airframe `4050_gz_x500_ps05`: `EKF2_GPS_CTRL=0`, `EKF2_EV_CTRL=3`, `EKF2_HGT_REF=3` (external vision), `UXRCE_DDS_SYNCT=0` (sim time), offboard-loss → LAND. `px4_interface` is the only node talking to PX4 (OffboardControlMode heartbeat, TrajectorySetpoint, VehicleCommand over uXRCE-DDS); if the autonomy stops sending setpoints the heartbeat stops and PX4's own failsafe takes over.

8. Gazebo architecture

World and vehicle are generated from YAML (`scripts/generate_world.py`, `generate_vehicle_model.py`): buildings, a long wall, poles, trees, an overhead canopy, a box cluster, guarded landing pad with a 0.40 m AprilTag 36h11, a tracking-station mast and three movable scenario objects. The X500 model has its NavSat sensor removed and four PS05 sensors added; an evaluation-only OdometryPublisher provides ground truth on `/sim/truth/odom`.

9. ROS 2 architecture

17 packages (16 ament_python + interfaces); every node uses sim time; configuration from `configs/*.yaml` with per-scenario overrides; JSON status topics for observability.

10. Sensor architecture

Forward RGB 640×480 @ 10 Hz (80°), 3D LiDAR 360×16 @ 10 Hz (±25°, 30 m), downward RGB 640×480 @ 15 Hz (60°), downward 32×32 flash LiDAR (70°×70°). Details: `docs/SENSOR_FUSION.md`.

11. Camera processing

`camera_processor` validates encoding/size/timestamps/CameraInfo, publishes health and applies scenario fault injection (blur, exposure, noise, timestamp offset, occlusion, blackout).

12. LiDAR processing

`lidar_processor`: finite/range/self filtering (0.6 m main, 0.35 m downward — the latter removes landing-gear returns), health, fault injection (Gaussian range noise, dropout, blackout).

13. TF2 architecture

world → map → odom → base_link → {camera, lidar, down_camera, down_lidar} links and REP-103 optical frames. Static mounts from `configs/sensors.yaml` (also used to generate the SDF; a unit test checks they agree); odom → base_link from EKF2. Runtime TF validation in `scripts/check_system.py`.

14. Timestamp synchronisation

Explicit nearest-neighbour pairing of each LiDAR scan with a camera frame, rejecting $|\Delta t| > 50$ ms; accepted/rejected counts and $|\Delta t|$ statistics published and logged.

15. Sensor fusion

$P_{cam} = T_{cam_lidar} \cdot P_{lidar}$ (TF2 at the scan stamp), $u = f_x \cdot X/Z + c_x$, $v = f_y \cdot Y/Z + c_y$ with the live CameraInfo; only in-image, in-front points are coloured; output `/fused/points` (x, y, z, rgb, in_fov). Projection-error metrics: extrinsic shift (px) and LiDAR-silhouette ↔ image-edge chamfer (px).

16. 3D mapping

Bounded 200×200×50 log-odds voxel map at 0.25 m with ray tracing (free space along each beam), FREE/UNKNOWN/OCCUPIED, fused RGB per voxel; degraded LiDAR-only mode if the fused stream stalls.

17. Obstacle inflation

Planning radius 1.20 m = vehicle 0.40 + localisation 0.15 + control 0.15 + clearance 0.25 + half planning cell 0.25 (`configs/safety.yaml`); live LiDAR hard envelope 0.70 m, kept below the planner's guaranteed surface clearance (0.77 m).

18. Approved mission corridor

Mission box $x, y \in [1, 49]$ m, $z \in [1, 10]$ m intersected with a 12 m tube around the approved centreline, minus a 0.5 m margin; A* cannot leave it and every trajectory is re-validated against it. PS05 spatial error E vs Emax = 2.0 m monitored in flight with alarm and containment.

19. 3D A*

26-connected weighted A* (w = 1.3, no corner cutting, optimistic unknown at 1.2× cost), line-of-sight pruning, verified Chaikin smoothing, trapezoidal speed profile (2 m/s, 1 m/s^2, 1 m/s vertical), single 2 cm validity check for pruning, smoothing and final validation. `docs/PLANNING.md`.

20. Local replanning

Look-ahead 8 m checked against every LiDAR scan and every map update; blocked → OBSTACLE_STOP (hover) → REPLANNING from the current pose inside the same corridor → NAVIGATE; no route → NO_SAFE_PATH (hover + 3 retries) → ABORTED.

21. GPS-denied design

No GNSS sensor on the airframe, GPS fusion disabled, external tracking (60 Hz, 10 ms latency, σ_r = 5 cm, $\sigma_\theta = \sigma_\varphi$ = 1.5 mrad) converted with the PS05 equations and fused by EKF2. Truth is evaluation-only; a bring-up finding (PX4 gz_bridge re-publishing the evaluation odometry as external vision) was fixed and is now guarded by static and runtime audit checks.

22. Landing target detection

AprilTag 36h11 (OpenCV ArUco), sub-pixel corners, IPPE-square PnP, TF2 to map, exponential filter; the marker is used only after GPS-free navigation reached the approximate landing zone.

23. LiDAR landing validation

Plane fit on the lowest returns in a 1.2 m disc around the tag: tilt ≤ 8°, RMS ≤ 5 cm, no returns > 0.15 m above the plane within 2.0 m, alignment ≤ 0.35 m. A marker alone never triggers descent.

24. Precision landing controller

PID per axis on the map-frame tag error (Kp 0.8, Ki 0.05, Kd 0.15, 0.6 m/s limit, clamped and conditional integration, filtered derivative); descent 0.4 → 0.25 m/s with re-validation at 2.0 / 1.0 / 0.5 m; blind-descent zone below 1.0 m; PX4 LAND at 0.45 m AGL.

25. Safety state machine

INIT → ARMING → TAKEOFF → PLANNING → NAVIGATE → LANDING_ZONE_SEARCH → TARGET_LOCK → ALIGN → DESCEND → FINAL_CHECK → LAND → LANDED, with OBSTACLE_STOP, REPLANNING, NO_SAFE_PATH, TARGET_LOST, UNSAFE_PAD, ABORT_CLIMB, RETURN_HOME, SENSOR_FAILURE, CORRIDOR_BREACH, ABORTED. Pure-Python, unit-tested against a kinematic vehicle for every branch. `docs/LANDING.md`.

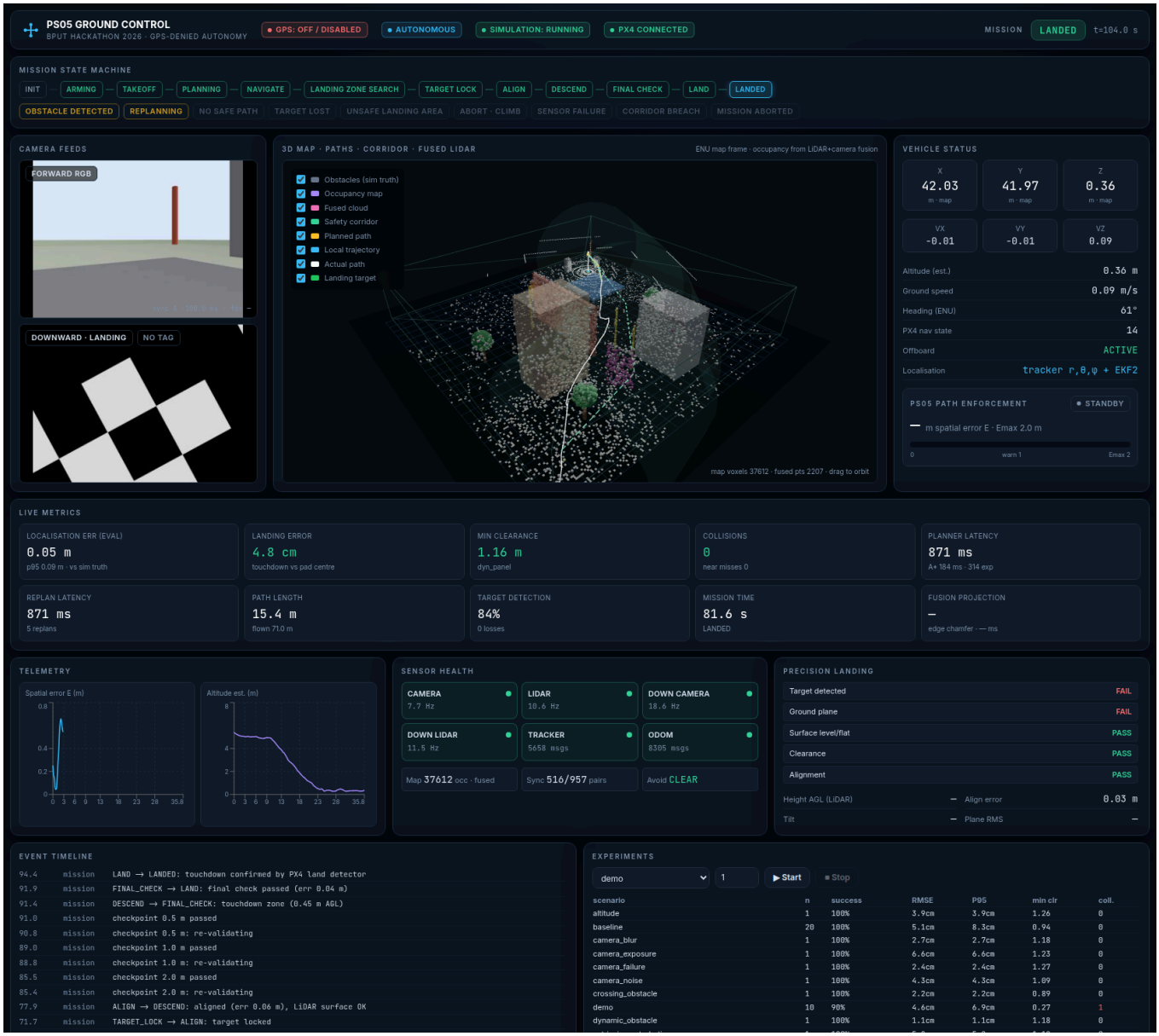
26. Backend architecture

FastAPI: REST endpoints for health, system, mission, vehicle, sensors, map, planner, landing, metrics (current + history), experiments (list/detail/start/stop), scenarios, logs; WebSocket `/ws/telemetry`; pydantic validation, CORS, structured logging; fed by the read-only `dashboard_bridge`.

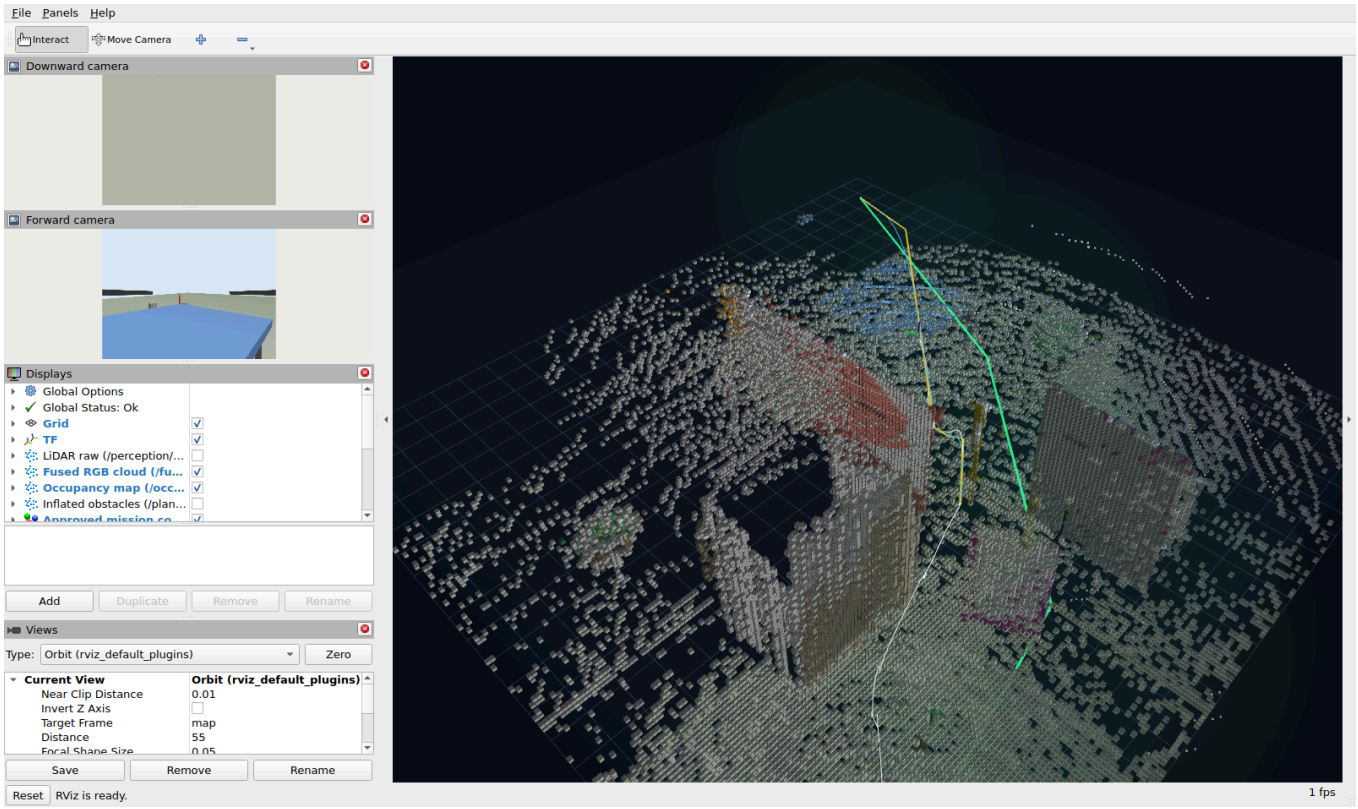
27. Frontend / dashboard

React + TypeScript + Vite + Tailwind + three.js: GPS/autonomy/simulation status, state-machine strip with failure states, forward and downward camera feeds (tag overlay), interactive 3D view (obstacles, occupancy voxels, fused cloud, corridor, planned/local/actual paths, target), vehicle status and PS05 E-vs-Emax gauge, live metrics, telemetry charts, sensor health, landing checklist, event timeline and experiment control/history. Only live or recorded data is displayed.

Ground-control dashboard during a live run (real telemetry):



RViz2 debugging view during a live run: fused-colour occupancy map, corridor, paths:



28. Test methodology

Level	Result
unit + regression tests	105 passed, 0 failed
backend tests	9 passed, 0 failed
static GPS / ground-truth audit	PASS
runtime GPS / ground-truth audit	PASS
integration system check (topics, CameraInfo, TF, PX4)	PASS
smoke flight arm → takeoff → hover → move → land	PASS; hover RMS error 0.094 m

Each end-to-end trial is a fresh launch of Gazebo, PX4 and all nodes (`scripts/run_mission.sh`) with a scenario file and seed; the evaluation node records `metrics.json`, `telemetry.csv` (10 Hz truth / estimate / reference), `events.jsonl`, `config.json` (all configs, config hash, git revision) and the PX4 ulog.

29. Experiment matrix

Scenario	Trials	Expected	Expectation met	Landed	Landing error mean	Landing RMSE	Collisions	Min clearance	Replans (mean)	Mission time (mean)
`altitude`	1	LANDED	100 %	1	3.9 cm	3.9 cm	0	1.26 m	2.0	74.8 s
`baseline`	23	LANDED	100 %	23	4.6 cm	5.3 cm	0	0.94 m	4.0	73.9 s
`camera_blur`	1	LANDED	100 %	1	2.7 cm	2.7 cm	0	1.18 m	10.0	98.9 s
`camera_exposure`	1	LANDED	100 %	1	6.6 cm	6.6 cm	0	1.23 m	4.0	72.9 s
`camera_failure`	1	LANDED	100 %	1	2.4 cm	2.4 cm	0	1.27 m	4.0	73.5 s
`camera_noise`	1	LANDED	100 %	1	4.3 cm	4.3 cm	0	1.08 m	5.0	75.4 s
`crossing_obstacle`	1	LANDED	100 %	1	2.2 cm	2.2 cm	0	1.34 m	6.0	77.2 s
`demo`	10	LANDED	100 %	10	4.4 cm	4.6 cm	0	1.11 m	6.6	84.2 s
`dynamic_obstacle`	1	LANDED	100 %	1	1.1 cm	1.1 cm	0	1.18 m	4.0	79.1 s
`extrinsic_perturbation`	1	LANDED	100 %	1	5.8 cm	5.8 cm	0	1.18 m	3.0	68.9 s
`landing_offset`	1	LANDED	100 %	1	4.6 cm	4.6 cm	0	1.13 m	4.0	84.9 s
`lidar_dropout`	1	LANDED	100 %	1	5.2 cm	5.2 cm	0	1.11 m	3.0	75.4 s

Scenario	Trials	Expected	Expectation met	Landed	Landing error mean	Landing RMSE	Collisions	Min clearance	Replans (mean)	Mission time (mean)
`lidar_failure`	1	LANDED	100 %	1	5.9 cm	5.9 cm	0	0.97 m	4.0	78.6 s
`lidar_noise`	1	LANDED	100 %	1	3.1 cm	3.1 cm	0	1.05 m	6.0	79.8 s
`narrow_pole`	1	LANDED	100 %	1	4.4 cm	4.4 cm	0	1.44 m	4.0	74.8 s
`no_path`	1	ABORTED	100 %	0	NOT MEASURED	NOT MEASURED	0	6.65 m	0.0	NOT MEASURED
`overhead`	1	LANDED	100 %	1	5.0 cm	5.0 cm	0	1.18 m	4.0	73.6 s
`pad_near_obstacle`	2	ABORTED	100 %	0	NOT MEASURED	NOT MEASURED	0	0.98 m	8.5	140.7 s
`pad_obstruction`	2	LANDED	100 %	2	4.1 cm	5.2 cm	0	1.12 m	5.0	91.8 s
`start_position`	3	LANDED	67 %	2	2.6 cm	2.8 cm	0	1.29 m	1.7	64.7 s
`start_yaw`	1	LANDED	100 %	1	7.5 cm	7.5 cm	0	1.06 m	4.0	72.5 s
`static_wall`	1	LANDED	100 %	1	4.8 cm	4.8 cm	0	1.16 m	5.0	81.6 s
`tag_loss`	1	LANDED	100 %	1	3.8 cm	3.8 cm	0	1.08 m	3.0	76.6 s
`timestamp_offset`	1	LANDED	100 %	1	4.1 cm	4.1 cm	0	1.09 m	5.0	74.2 s
`timestamp_offset_small`	1	LANDED	100 %	1	3.9 cm	3.9 cm	0	1.48 m	4.0	71.4 s
`tracker_noise`	1	LANDED	100 %	1	9.8 cm	9.8 cm	0	1.38 m	5.0	78.5 s

30. Metrics

Landing error $E = \sqrt{((Px-Tx)^2+(Py-Ty)^2)}$ at PX4-detected touchdown (truth) vs. pad centre; localisation error = |EKF estimate − truth|; true spatial error = |truth − trajectory reference| during NAVIGATE (PS05 E); minimum clearance = distance from the vehicle centre to the nearest obstacle surface (truth geometry); collision = clearance < vehicle radius 0.40 m; near miss = clearance < 0.65 m; planning / replanning latency = planner service time; detection latency = unexpected-obstacle placement → first detection; map accuracy = occupied voxels farther than 0.5 m from any true surface (phantoms).

31. Results

Baseline (static course)

Metric	Value
Trials	23
Landed / success rate	23 / 100 %
Landing error mean / median	4.6 cm / 3.4 cm
Landing error RMSE / P95 / max	5.3 cm / 8.2 cm / 9.3 cm
Collisions (total) / near misses	0 / 0
Minimum clearance (worst trial)	0.94 m
Executed path length (mean)	69.1 m
Initial planned path length (mean)	53.3 m
Planning latency mean / P95	1126 ms / 1952 ms
Replanning latency mean / P95	940 ms / 1337 ms
Replans per mission (mean)	4.0
Stop response (detection → hover, mean)	1.26 s
Target detection rate during landing (mean)	83 %
Localisation error (mean of run means)	0.038 m
True spatial error E during NAVIGATE (mean of run means)	0.343 m
True spatial error E (worst run max)	1.01 m
Fusion edge-alignment error (mean)	7.9 px
Camera–LiDAR sync abs(Δt) (mean)	0.0 ms
Mission time (mean)	73.9 s

Dynamic obstacle (demo)

Metric	Value
Trials	10
Landed / success rate	10 / 100 %
Landing error mean / median	4.4 cm / 4.2 cm
Landing error RMSE / P95 / max	4.6 cm / 6.9 cm / 8.0 cm
Collisions (total) / near misses	0 / 0
Minimum clearance (worst trial)	1.11 m
Executed path length (mean)	74.9 m
Initial planned path length (mean)	53.0 m
Planning latency mean / P95	1188 ms / 1786 ms
Replanning latency mean / P95	1098 ms / 1482 ms
Replans per mission (mean)	6.6
Stop response (detection → hover, mean)	1.13 s
Target detection rate during landing (mean)	80 %
Localisation error (mean of run means)	0.034 m
True spatial error E during NAVIGATE (mean of run means)	0.291 m
True spatial error E (worst run max)	1.06 m
Fusion edge-alignment error (mean)	8.5 px
Camera–LiDAR sync abs(Δt) (mean)	0.0 ms
Mission time (mean)	84.2 s

Baseline by code revision (the earlier revision differs only in failure-handling branches that none of its runs entered; "-dirty" flags on those runs come from uncommitted documentation edits):

Revision	Trials	Landed	Landing error mean	RMSE	Collisions	Min clearance
`75d4b5d`	3	3	6.0 cm	6.3 cm	0	1.00 m
`18eba4c`	20	20	4.4 cm	5.1 cm	0	0.94 m

Unexpected-obstacle detection latency (demo runs, sim time): mean 1.51 s, max 3.50 s (n = 10).

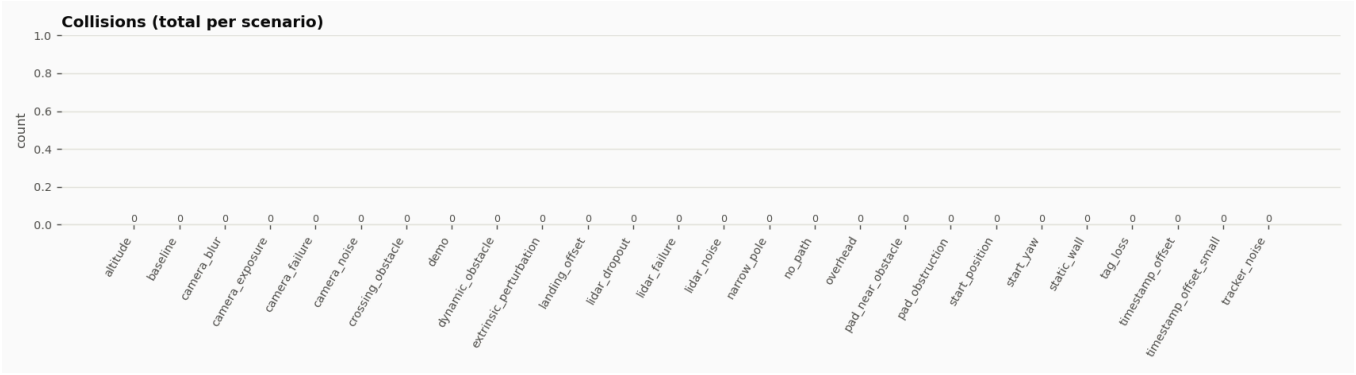
Clearance, collisions and near misses were re-evaluated for all 61 runs with yaw-aware world geometry (`scripts/reevaluate_clearance.py`, from each run's logged truth telemetry and scenario events); evaluator v1 had modelled rotated scenario objects as axis-aligned boxes. Changed runs: `eval_dynamic_t04` (min clearance 0.51 → 1.29 m, collisions 0 → 0), `eval_dynamic_t06` (min clearance 0.56 → 1.19 m, collisions 0 → 0), `eval_dynamic_t07` (min clearance 0.82 → 1.27 m, collisions 0 → 0), `eval_dynamic_t08` (min clearance 0.85 → 1.29 m, collisions 0 → 0), `eval_dynamic_t09` (min clearance 0.66 → 1.22 m, collisions 0 → 0), `eval_dynamic_t10` (min clearance 0.27 → 1.29 m, collisions 1 → 0), `stress_crossing_obstacle_t01` (min clearance 0.89 → 1.34 m, collisions 0 → 0). Original values remain in each `metrics.json` under `evaluator_v1_*`.

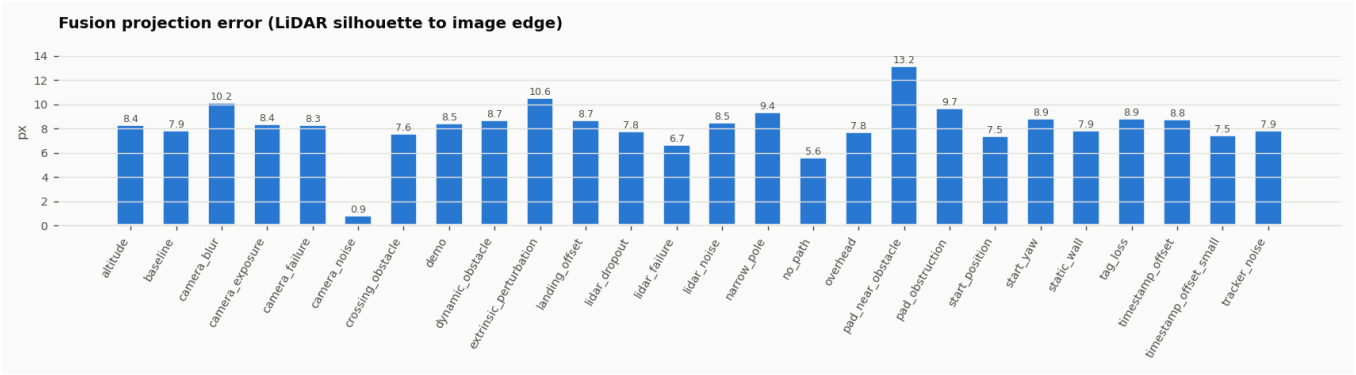
Occupancy-map accuracy across runs: phantom voxel fraction mean 0.79 %, max 3.71 %; median voxel-to-surface error 0.125 m.

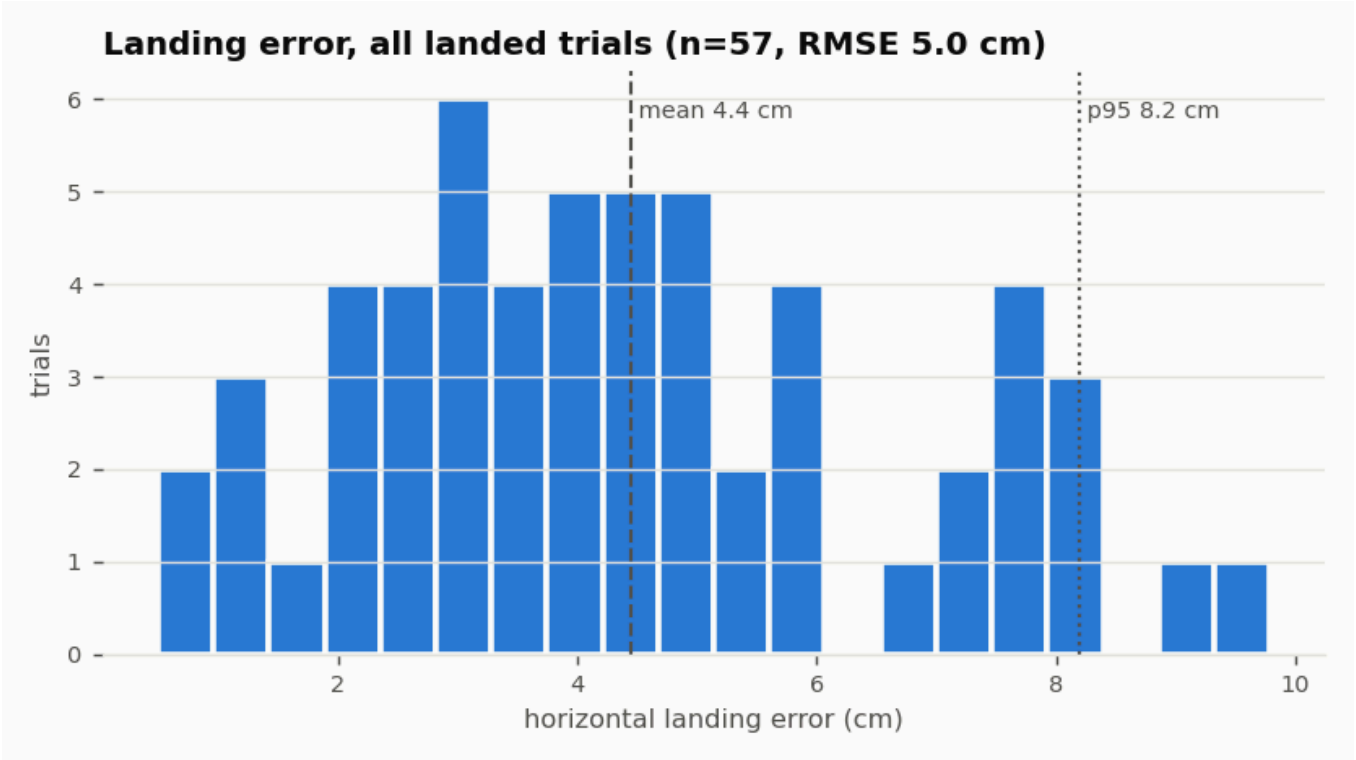
Planner benchmark (`scripts/bench_planner.py`, mission query with far space unknown): {"rasterize_s": 11.88011465800082, "grid_build_ms": 127.44095299967739, "corridor_mask_ms": 102.32211299990013, "astar_ms": 137.7257419990201, "trajectory_ms": 4.004734000773169, "success": true, "expansions": 2294, "length_m": 53.727979910661446, "waypoints": 6, "traj_duration_s": 30.776619294369674, "grid_shape": [100, 100, 25]}

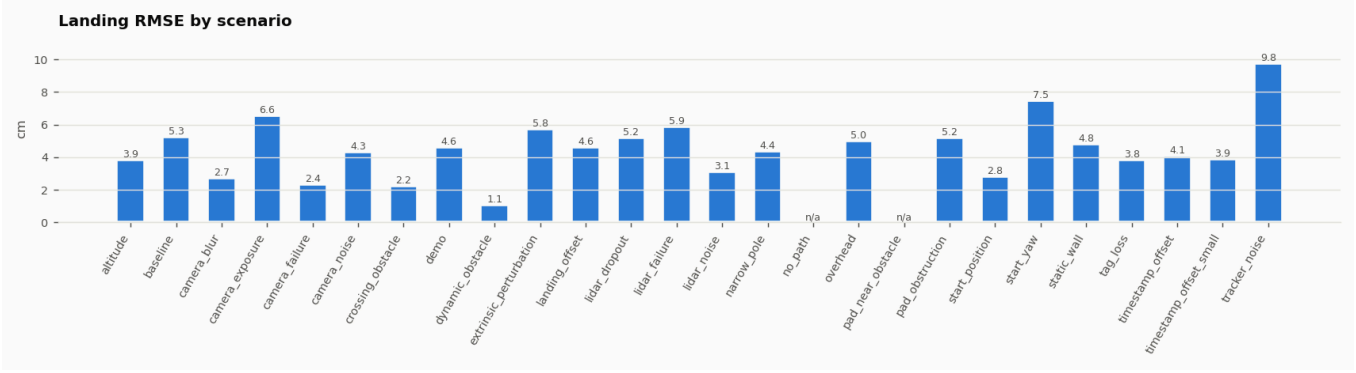
PS05 Phase-1 target "mean simulated trajectory offset E < 0.5 m": baseline mean of run means 0.343 m (MET).

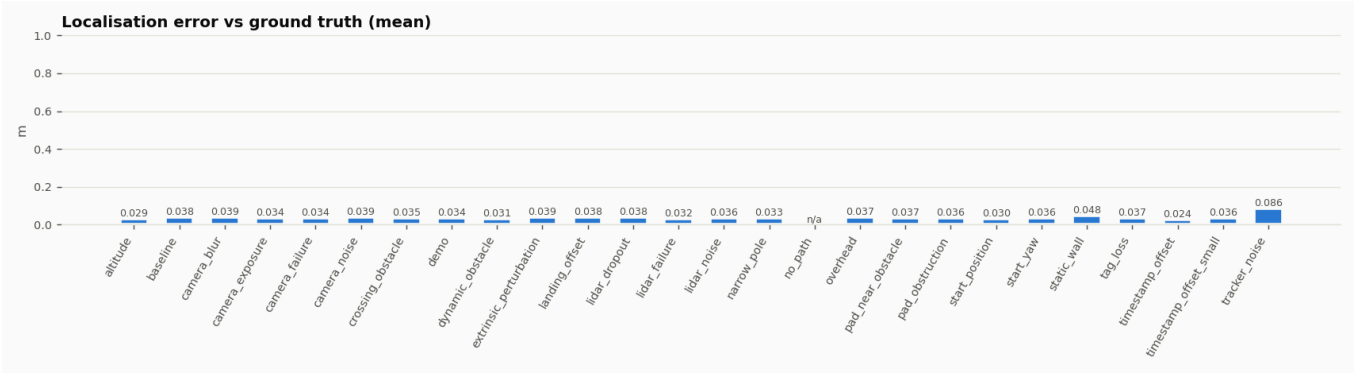
Figures (in `results/plots/`): landing-error distribution, touchdown scatter, per-scenario landing RMSE, success rate, minimum clearance, collision count, planning/replanning latency, path length, target detection rate, fusion projection error, localisation error, and planned-vs-actual trajectory + spatial-error time series.

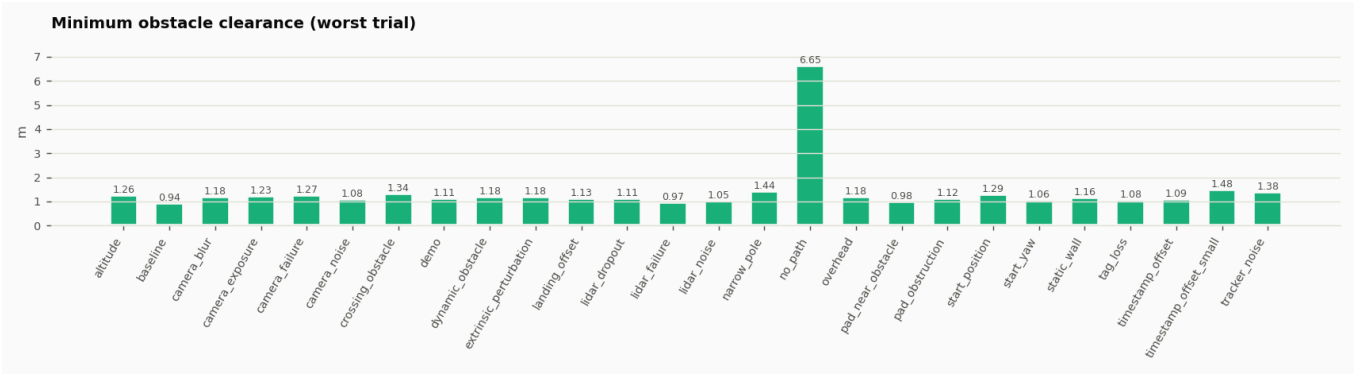

Collisions (total per scenario)
count
1.0
0.8
0.6
0.4
0.2
0.0
0
altitude
baseline
camera_blur
camera_exposure
camera_failure
camera_noise
crossing_obstacle
demo
dynamic_obstacle
extrinsic_perturbation
landing_offset
lidar_dropout
lidar_failure
lidar_noise
narrow_pole
no_path
overhead
pad_near_obstacle
pad_obstruction
start_position
start_yaw
static_wall
tag_loss
timestamp_offset
timestamp_offset_small
tracker_noise

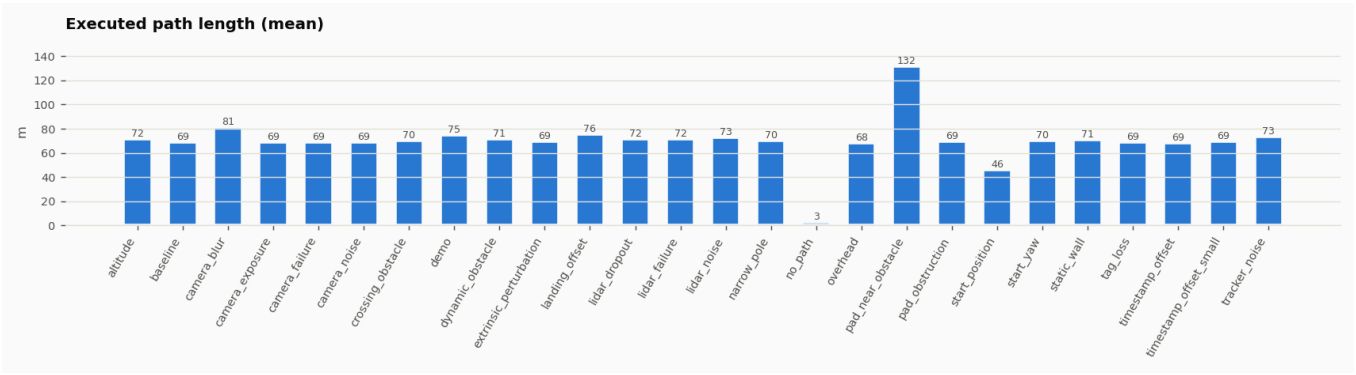

Fusion projection error (LiDAR silhouette to image edge)
px
14
12
10
8
6
4
2
0
8.4
7.9
10.2
8.4
8.3
0.9
7.6
8.5
8.7
10.6
8.7
7.8
6.7
8.5
9.4
5.6
7.8
13.2
9.7
7.5
8.9
7.9
8.9
8.8
7.5
7.9
altitude
baseline
camera_blur
camera_exposure
camera_failure
camera_noise
crossing_obstacle
demo
dynamic_obstacle
extrinsic_perturbation
landing_offset
lidar_dropout
lidar_failure
lidar_noise
narrow_pole
no_path
overhead
pad_near_obstacle
pad_obstruction
start_position
start_yaw
static_wall
tag_loss
timestamp_offset
timestamp_offset_small
tracker_noise

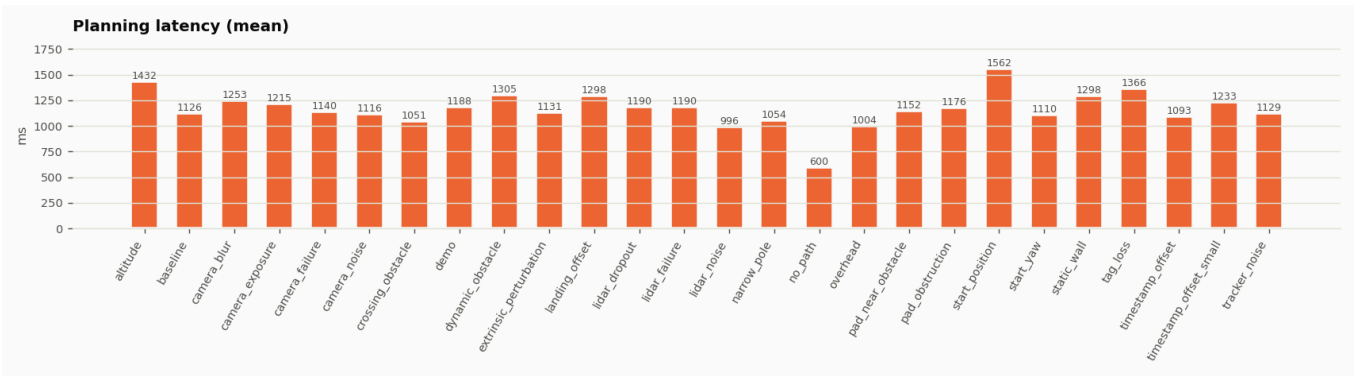

Landing error, all landed trials (n=57, RMSE 5.0 cm)
mean 4.4 cm
p95 8.2 cm
trials
6
5
4
3
2
1
0
2
4
6
8
10
horizontal landing error (cm)

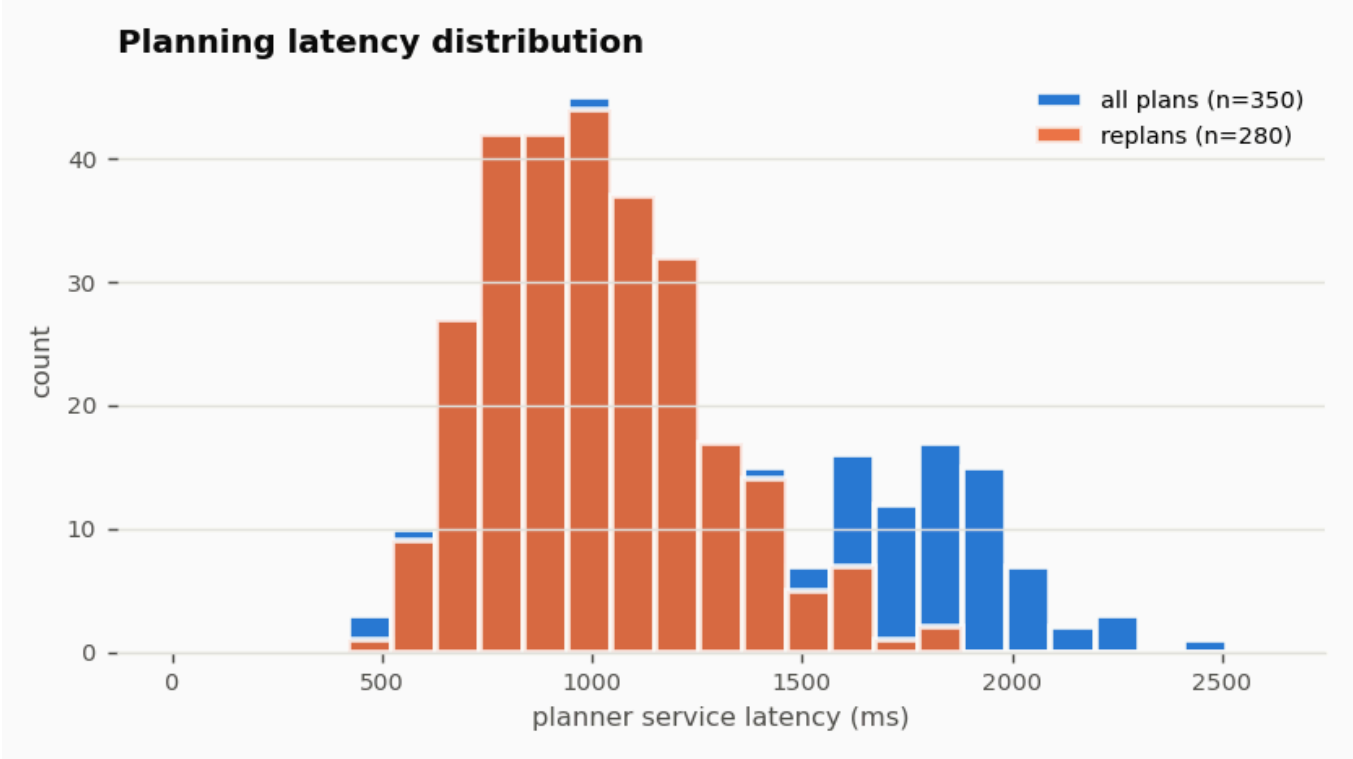

Landing RMSE by scenario
cm
10
8
6
4
2
0
3.9
5.3
2.7
6.6
2.4
4.3
2.2
4.6
1.1
5.8
4.6
5.2
5.9
3.1
4.4
n/a
5.0
n/a
5.2
2.8
7.5
4.8
3.8
4.1
3.9
9.8
altitude
baseline
camera_blur
camera_exposure
camera_failure
camera_noise
crossing_obstacle
demo
dynamic_obstacle
extrinsic_perturbation
landing_offset
lidar_dropout
lidar_failure
lidar_noise
narrow_pole
no_path
overhead
pad_near_obstacle
pad_obstruction
start_position
start_yaw
static_wall
tag_loss
timestamp_offset
timestamp_offset_small
tracker_noise

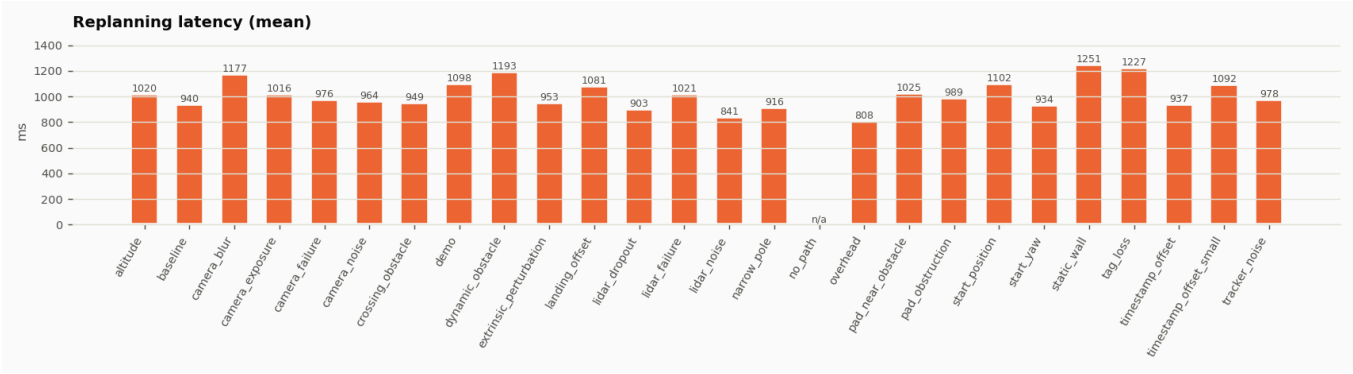

Localisation error vs ground truth (mean)
m
1.0
0.8
0.6
0.4
0.2
0.0
0.029
0.038
0.039
0.034
0.034
0.039
0.035
0.034
0.031
0.039
0.038
0.038
0.032
0.036
0.033
n/a
0.037
0.037
0.036
0.030
0.036
0.048
0.037
0.024
0.036
0.086
altitude
baseline
camera_blur
camera_exposure
camera_failure
camera_noise
crossing_obstacle
demo
dynamic_obstacle
extrinsic_perturbation
landing_offset
lidar_dropout
lidar_failure
lidar_noise
narrow_pole
no_path
overhead
pad_near_obstacle
pad_obstruction
start_position
start_yaw
static_wall
tag_loss
timestamp_offset
timestamp_offset_small
tracker_noise

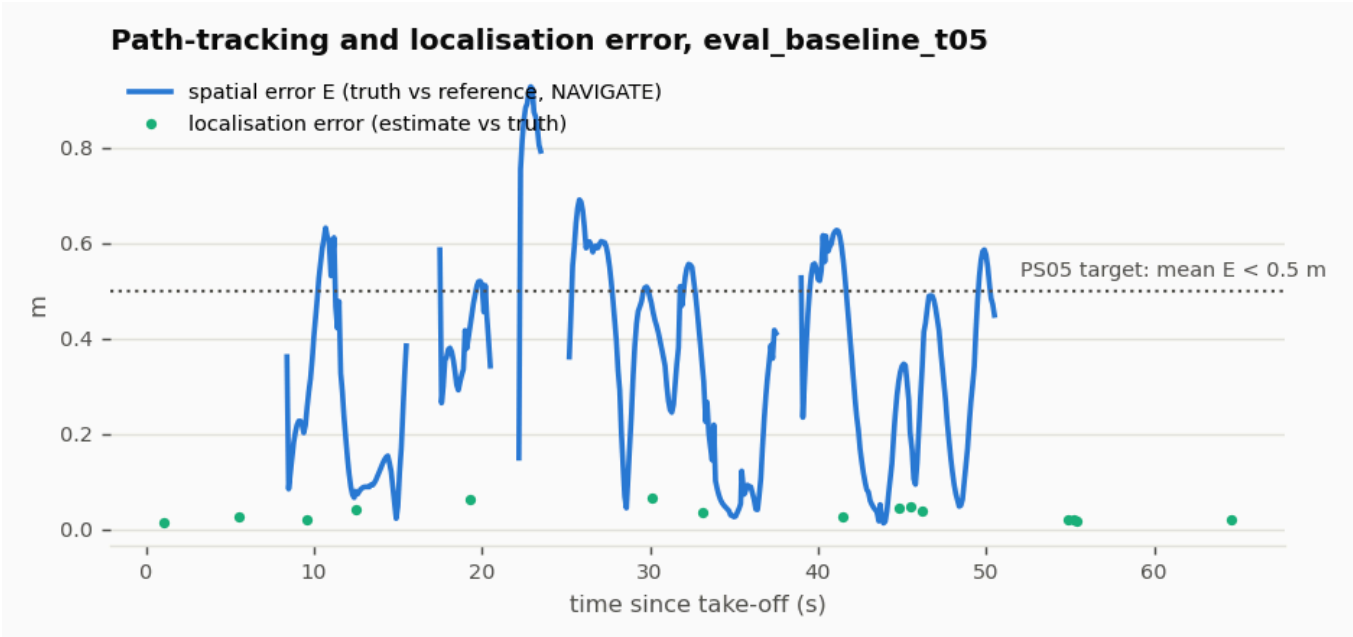

Minimum obstacle clearance (worst trial)
m
7
6
5
4
3
2
1
0
1.26
0.94
1.18
1.23
1.27
1.08
1.34
1.11
1.18
1.18
1.13
1.11
0.97
1.05
1.44
6.65
1.18
0.98
1.12
1.29
1.06
1.16
1.08
1.09
1.48
1.38
altitude
baseline
camera_blur
camera_exposure
camera_failure
camera_noise
crossing_obstacle
demo
dynamic_obstacle
extrinsic_perturbation
landing_offset
lidar_dropout
lidar_failure
lidar_noise
narrow_pole
no_path
overhead
pad_near_obstacle
pad_obstruction
start_position
start_yaw
static_wall
tag_loss
timestamp_offset
timestamp_offset_small
tracker_noise

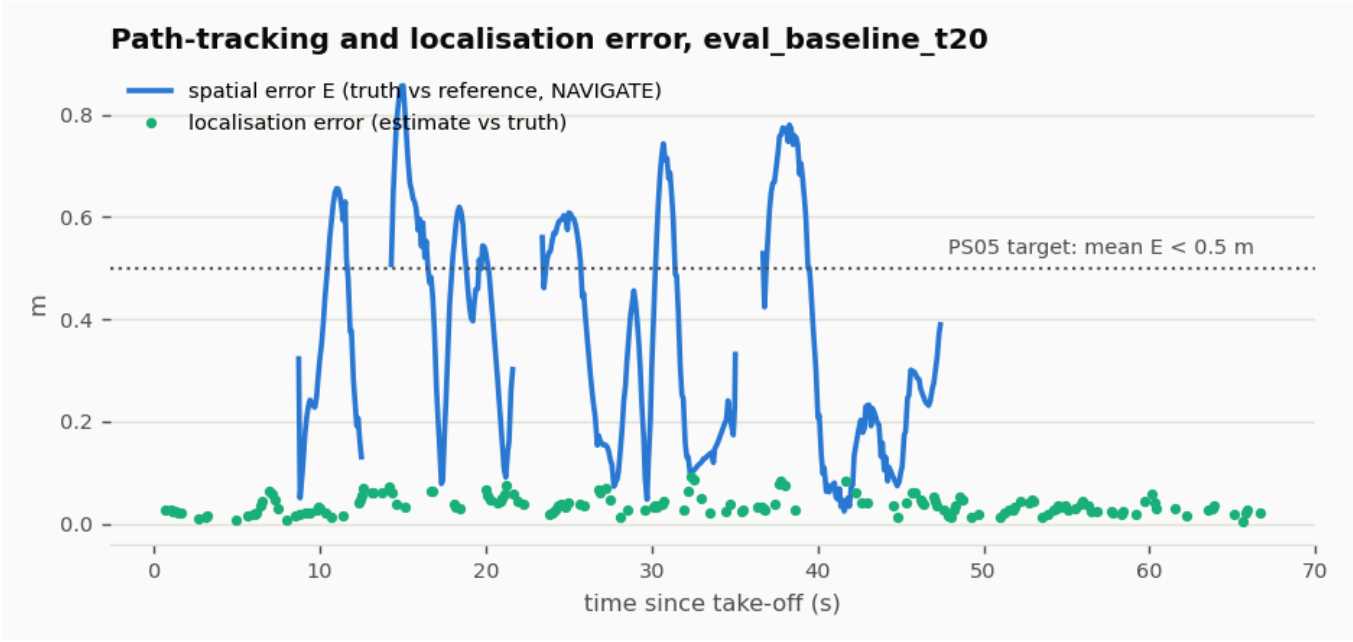

Executed path length (mean)
m
140
120
100
80
60
40
20
0
72
69
81
69
69
69
70
75
71
69
76
72
72
73
70
3
68
132
69
46
70
71
69
69
69
73
altitude
baseline
camera_blur
camera_exposure
camera_failure
camera_noise
crossing_obstacle
demo
dynamic_obstacle
extrinsic_perturbation
landing_offset
lidar_dropout
lidar_failure
lidar_noise
narrow_pole
no_path
overhead
pad_near_obstacle
pad_obstruction
start_position
start_yaw
static_wall
tag_loss
timestamp_offset
timestamp_offset_small
tracker_noise

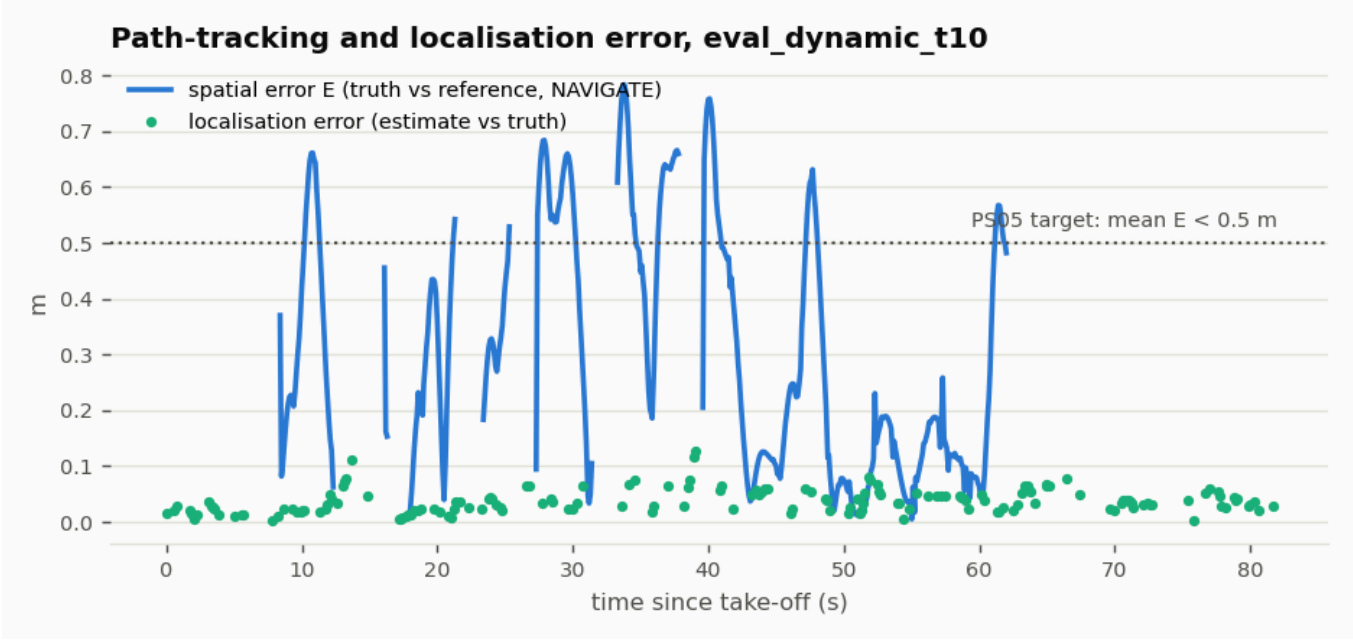

Planning latency (mean)
ms
1750
1500
1250
1000
750
500
250
0
1432
1126
1253
1215
1140
1116
1051
1188
1305
1131
1298
1190
1190
996
1054
600
1004
1152
1176
1562
1110
1298
1366
1093
1233
1129
altitude
baseline
camera_blur
camera_exposure
camera_failure
camera_noise
crossing_obstacle
demo
dynamic_obstacle
extrinsic_perturbation
landing_offset
lidar_dropout
lidar_failure
lidar_noise
narrow_pole
no_path
overhead
pad_near_obstacle
pad_obstruction
start_position
start_yaw
static_wall
tag_loss
timestamp_offset
timestamp_offset_small
tracker_noise

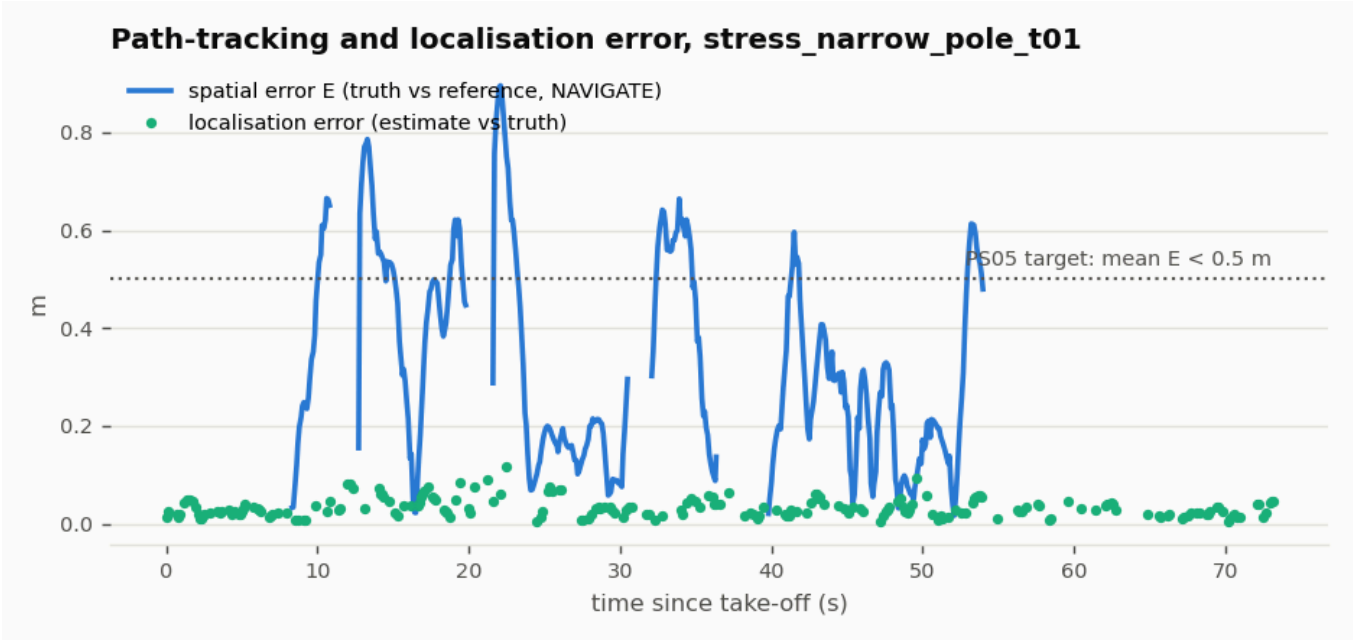

Planning latency distribution
all plans (n=350)
replans (n=280)
count
planner service latency (ms)
0
10
20
30
40
0
500
1000
1500
2000
2500

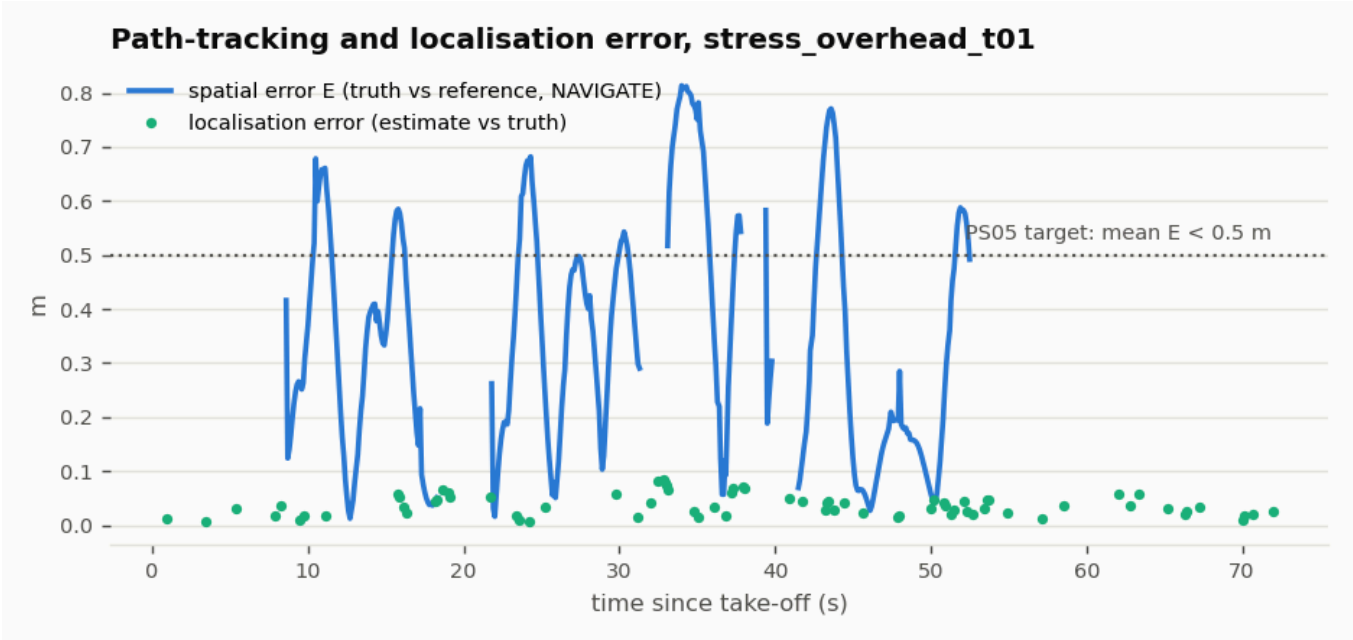

Replanning latency (mean)
ms
1020
940
1177
1016
976
964
949
1098
1193
953
1081
903
1021
841
916
n/a
808
1025
989
1102
934
1251
1227
937
1092
978
altitude
baseline
camera_blur
camera_exposure
camera_failure
camera_noise
crossing_obstacle
demo
dynamic_obstacle
extrinsic_perturbation
landing_offset
lidar_dropout
lidar_failure
lidar_noise
narrow_pole
no_path
overhead
pad_near_obstacle
pad_obstruction
start_position
start_yaw
static_wall
tag_loss
timestamp_offset
timestamp_offset_small
tracker_noise

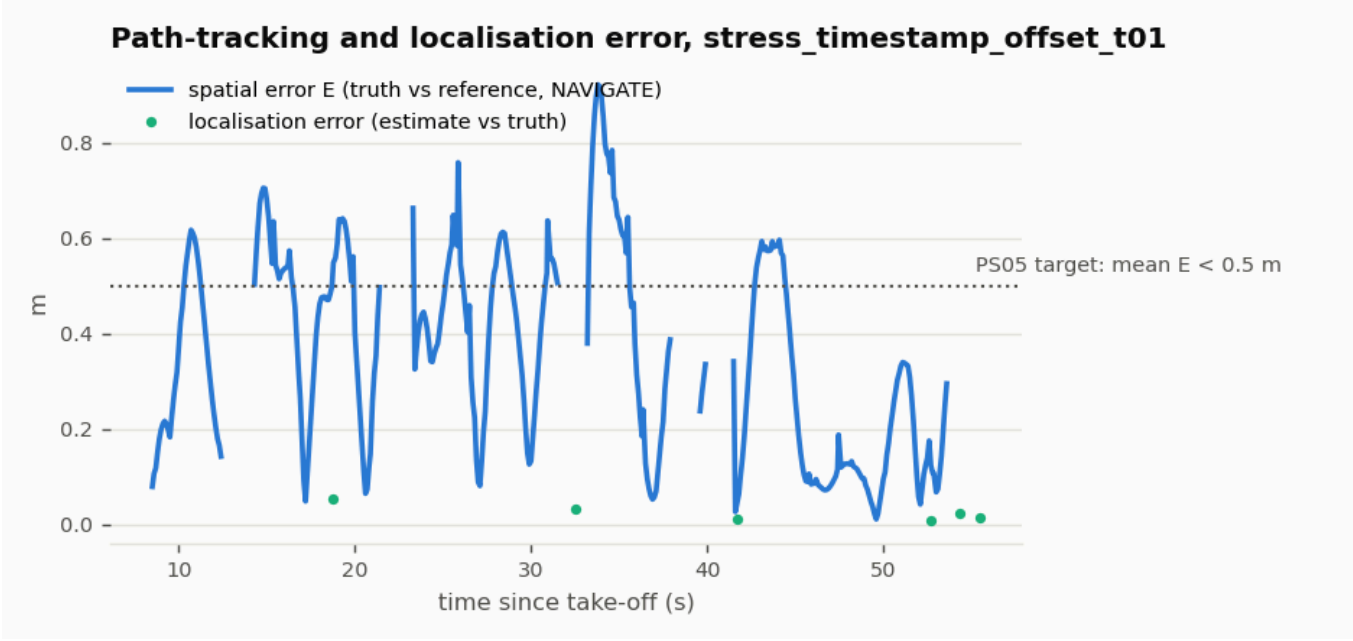

Path-tracking and localisation error, eval_baseline_t05
spatial error E (truth vs reference, NAVIGATE)
localisation error (estimate vs truth)
PS05 target: mean E < 0.5 m
m
time since take-off (s)
0.0
0.2
0.4
0.6
0.8
0
10
20
30
40
50
60

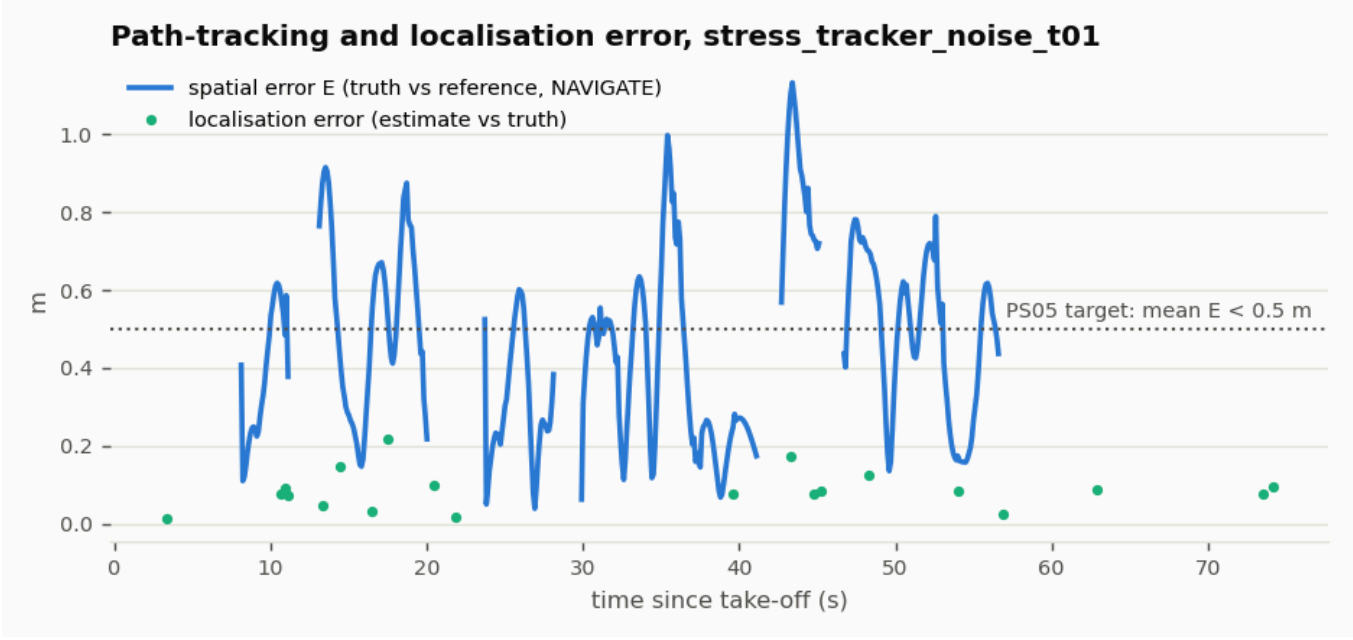

Path-tracking and localisation error, eval_baseline_t20
spatial error E (truth vs reference, NAVIGATE)
localisation error (estimate vs truth)
PS05 target: mean E < 0.5 m
m
time since take-off (s)

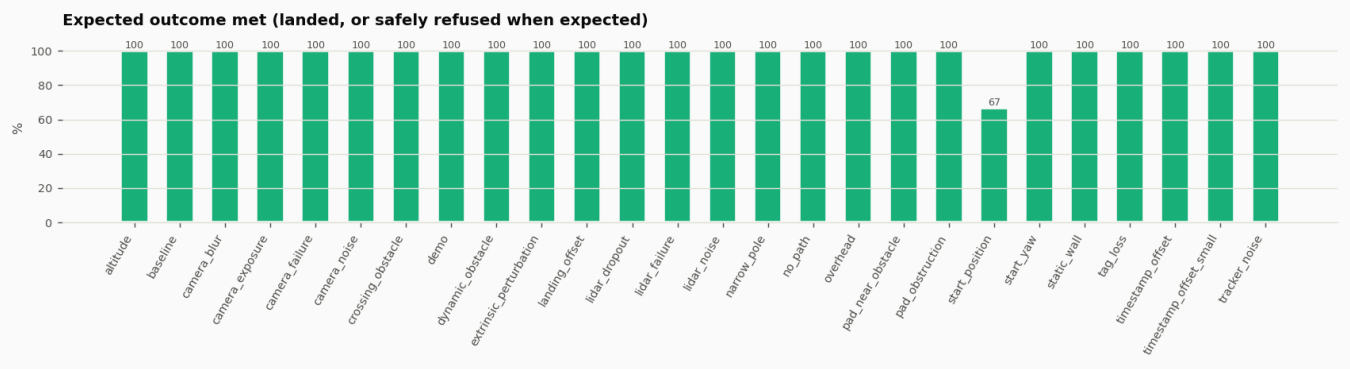

Path-tracking and localisation error, eval_dynamic_t10
spatial error E (truth vs reference, NAVIGATE)
localisation error (estimate vs truth)
PS05 target: mean E < 0.5 m
m
time since take-off (s)

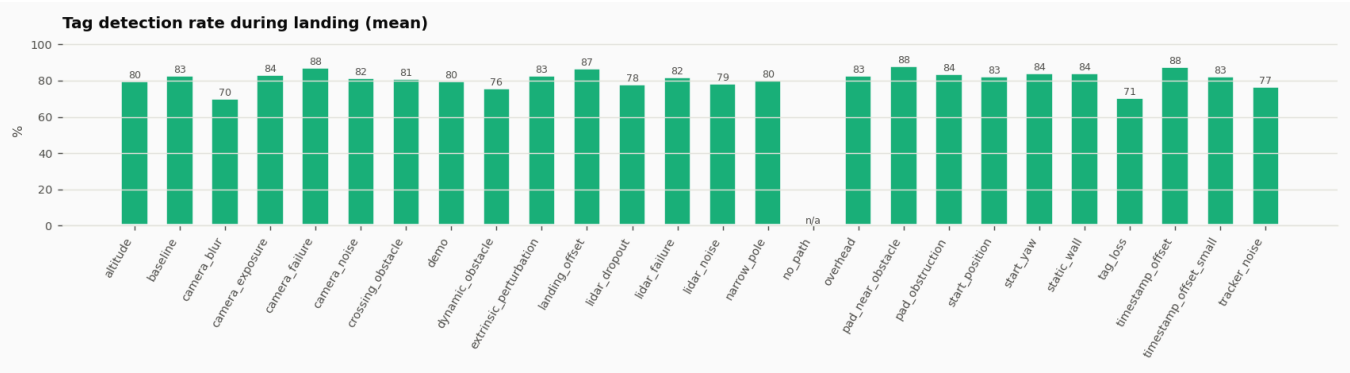

Path-tracking and localisation error, stress_narrow_pole_t01
spatial error E (truth vs reference, NAVIGATE)
localisation error (estimate vs truth)
PS05 target: mean E < 0.5 m
m
time since take-off (s)

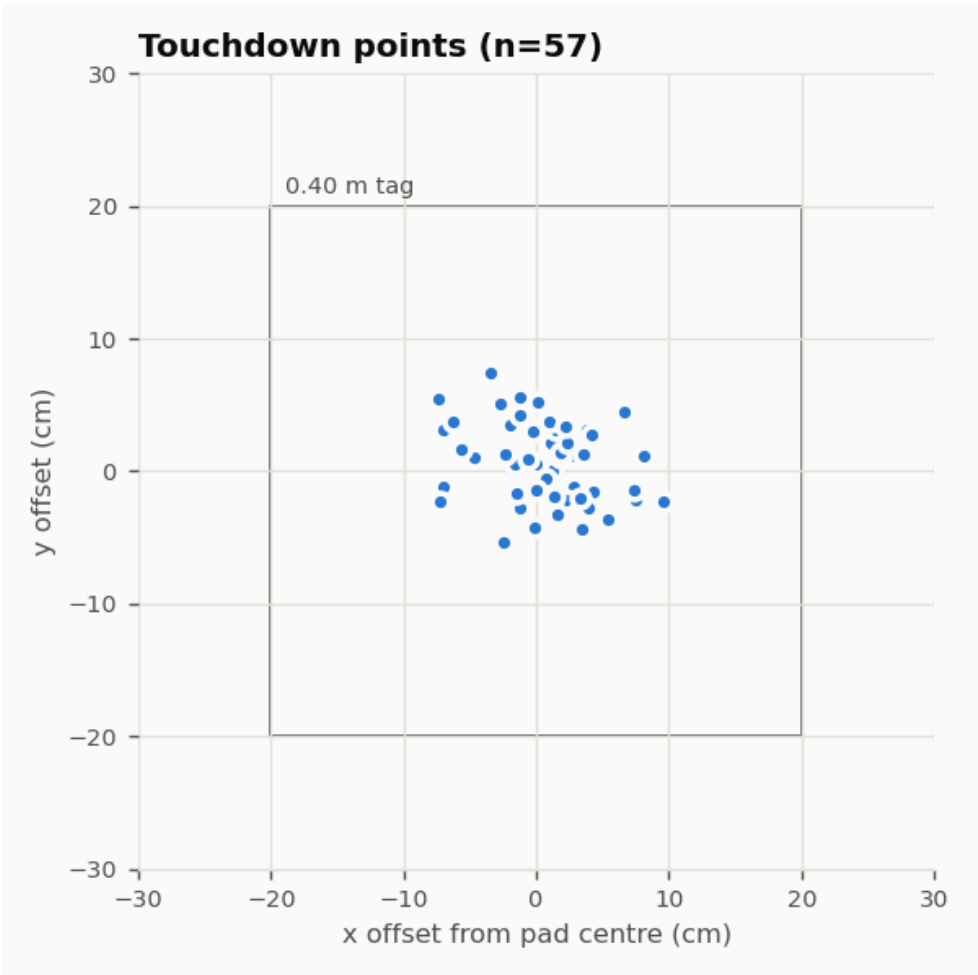

Path-tracking and localisation error, stress_overhead_t01
spatial error E (truth vs reference, NAVIGATE)
localisation error (estimate vs truth)
PS05 target: mean E < 0.5 m
m
time since take-off (s)

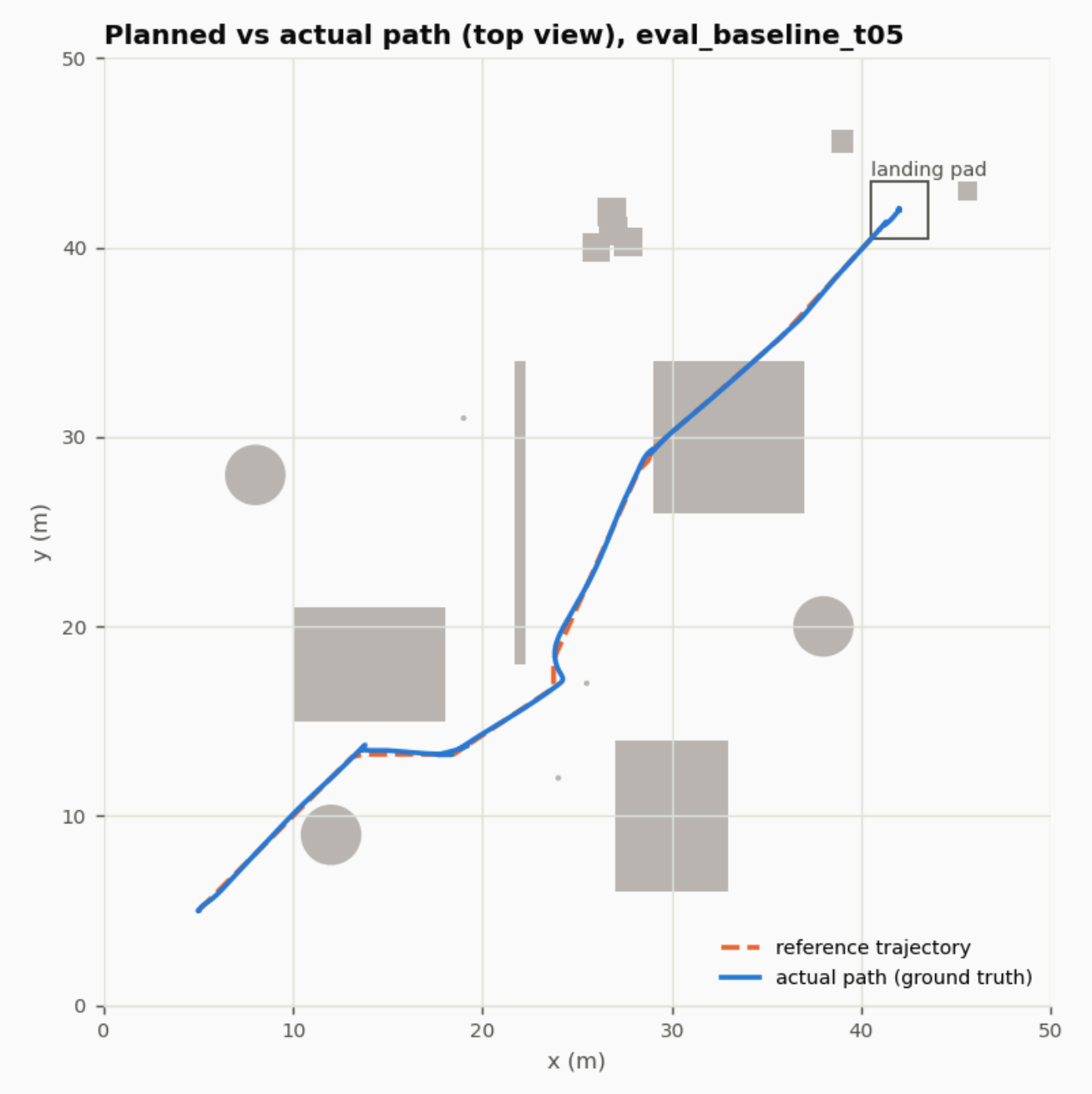

Path-tracking and localisation error, stress_timestamp_offset_t01
spatial error E (truth vs reference, NAVIGATE)
localisation error (estimate vs truth)
PS05 target: mean E < 0.5 m
m
time since take-off (s)

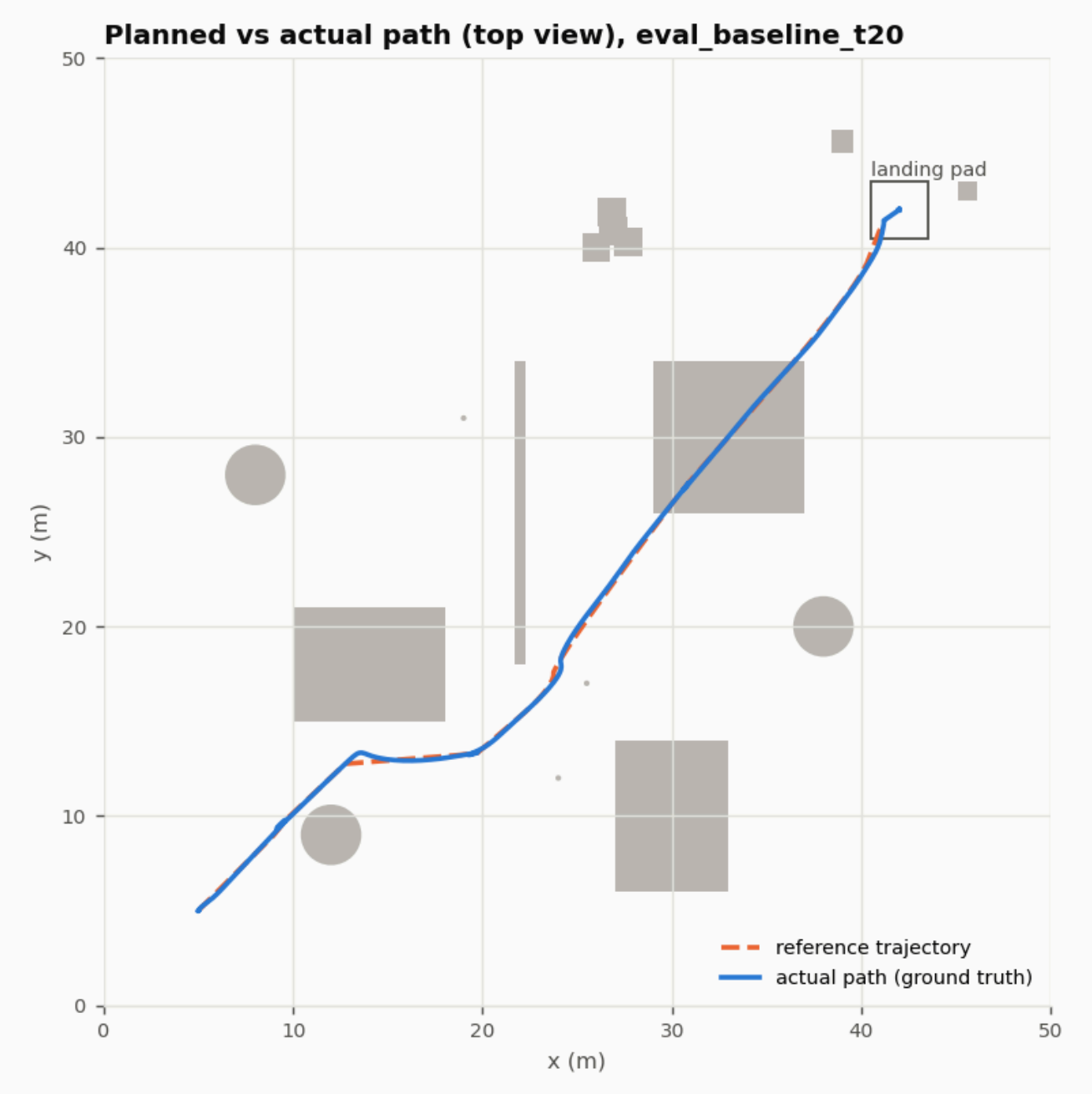

Path-tracking and localisation error, stress_tracker_noise_t01
spatial error E (truth vs reference, NAVIGATE)
localisation error (estimate vs truth)
PS05 target: mean E < 0.5 m
m
time since take-off (s)

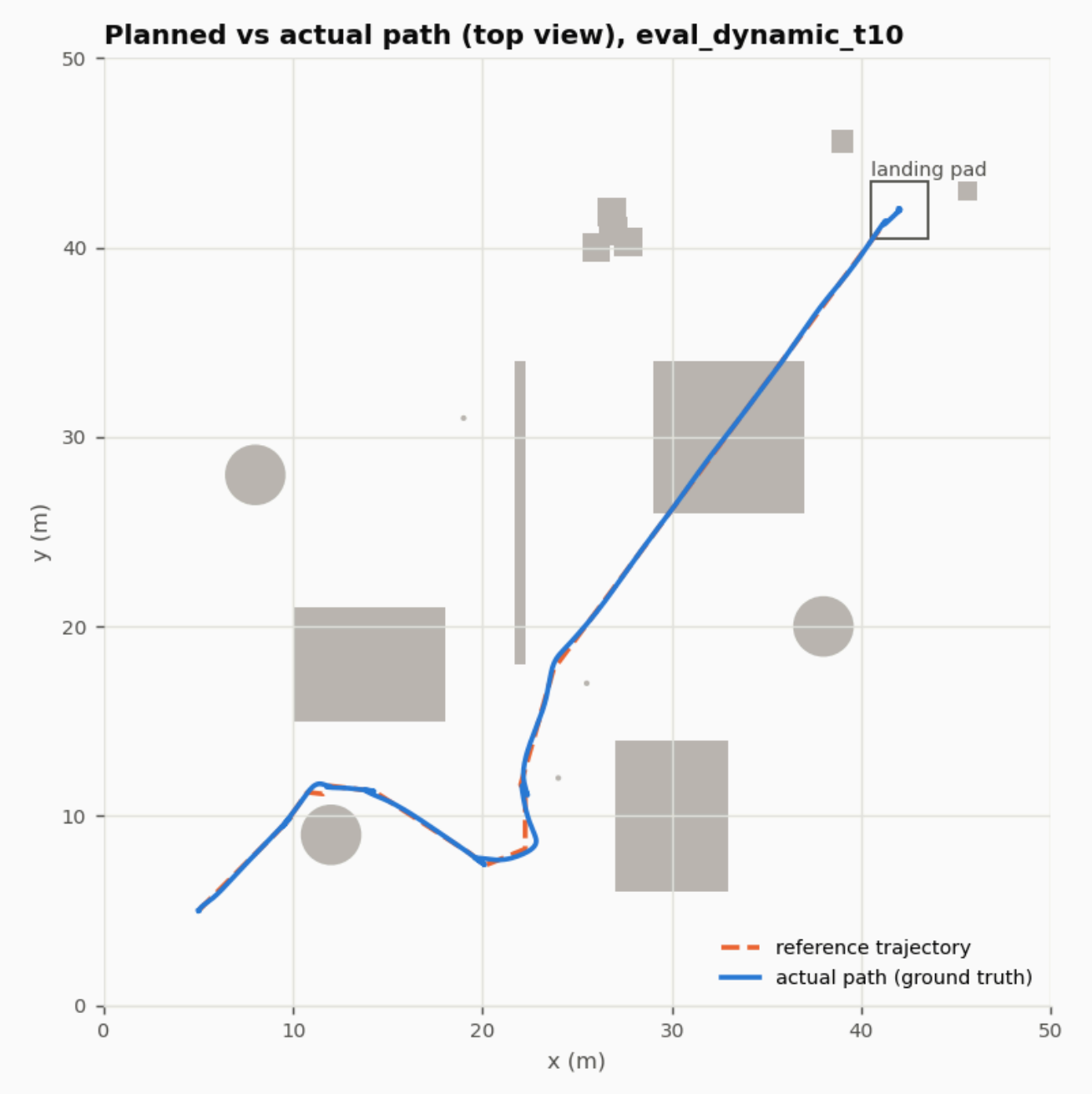

Expected outcome met (landed, or safely refused when expected)
%
100
80
60
40
20
0
altitude 100
baseline 100
camera_blur 100
camera_exposure 100
camera_failure 100
camera_noise 100
crossing_obstacle 100
demo 100
dynamic_obstacle 100
extrinsic_perturbation 100
landing_offset 100
lidar_dropout 100
lidar_failure 100
lidar_noise 100
narrow_pole 100
no_path 100
overhead 100
pad_near_obstacle 100
pad_obstruction 100
start_position 67
start_yaw 100
static_wall 100
tag_loss 100
timestamp_offset 100
timestamp_offset_small 100
tracker_noise 100

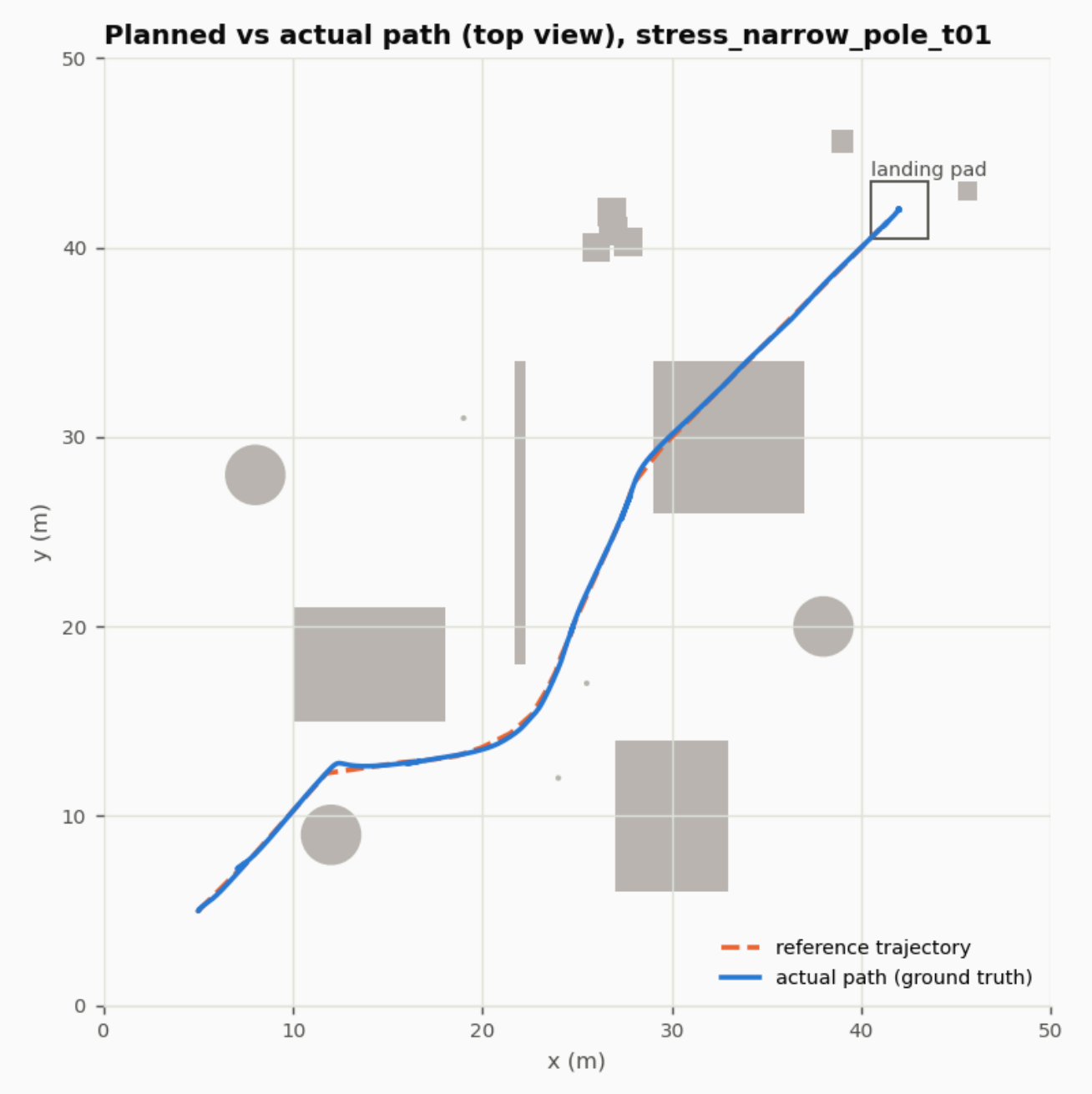

Tag detection rate during landing (mean)
%
100
80
60
40
20
0
altitude 80
baseline 83
camera_blur 70
camera_exposure 84
camera_failure 88
camera_noise 82
crossing_obstacle 81
demo 80
dynamic_obstacle 76
extrinsic_perturbation 83
landing_offset 87
lidar_dropout 78
lidar_failure 82
lidar_noise 79
narrow_pole 80
no_path n/a
overhead 83
pad_near_obstacle 88
pad_obstruction 84
start_position 83
start_yaw 84
static_wall 84
tag_loss 71
timestamp_offset 88
timestamp_offset_small 83
tracker_noise 77

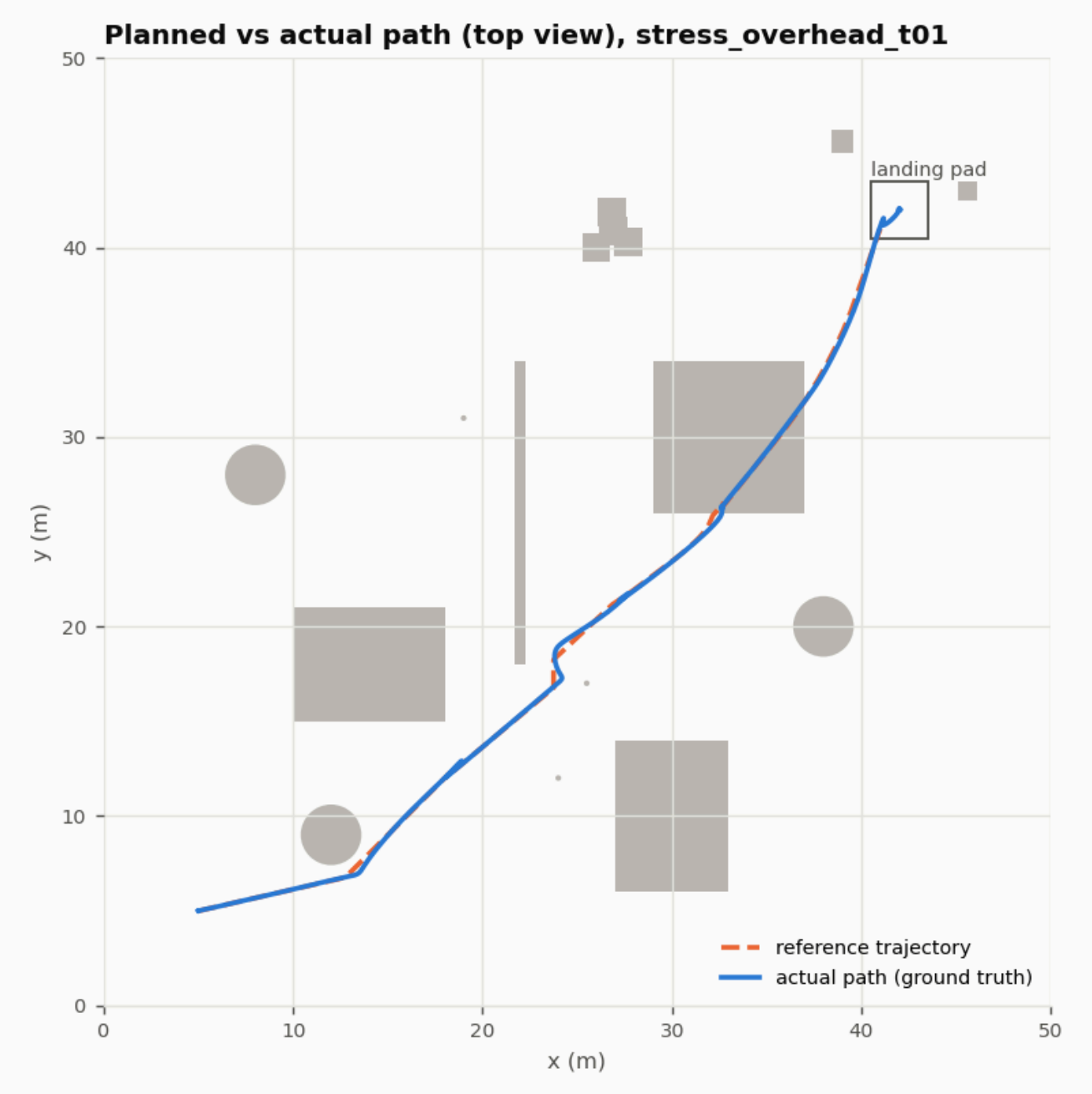

Touchdown points (n=57)
0.40 m tag
y offset (cm)
30
20
10
0
−10
−20
−30
−30
−20
−10
0
10
20
30
x offset from pad centre (cm)

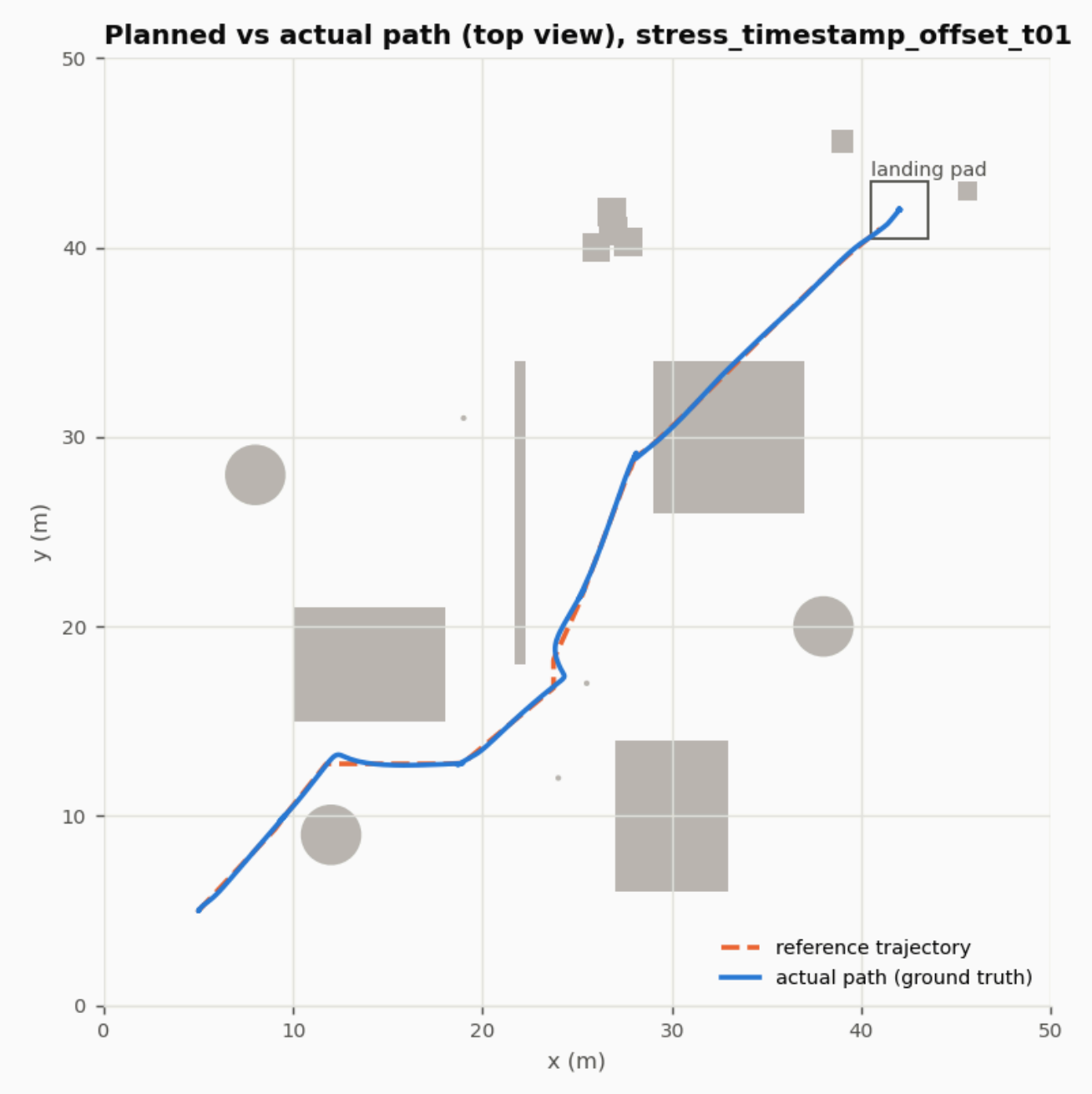

Planned vs actual path (top view), eval_baseline_t05
landing pad
reference trajectory
actual path (ground truth)
x (m)
y (m)
0
10
20
30
40
50

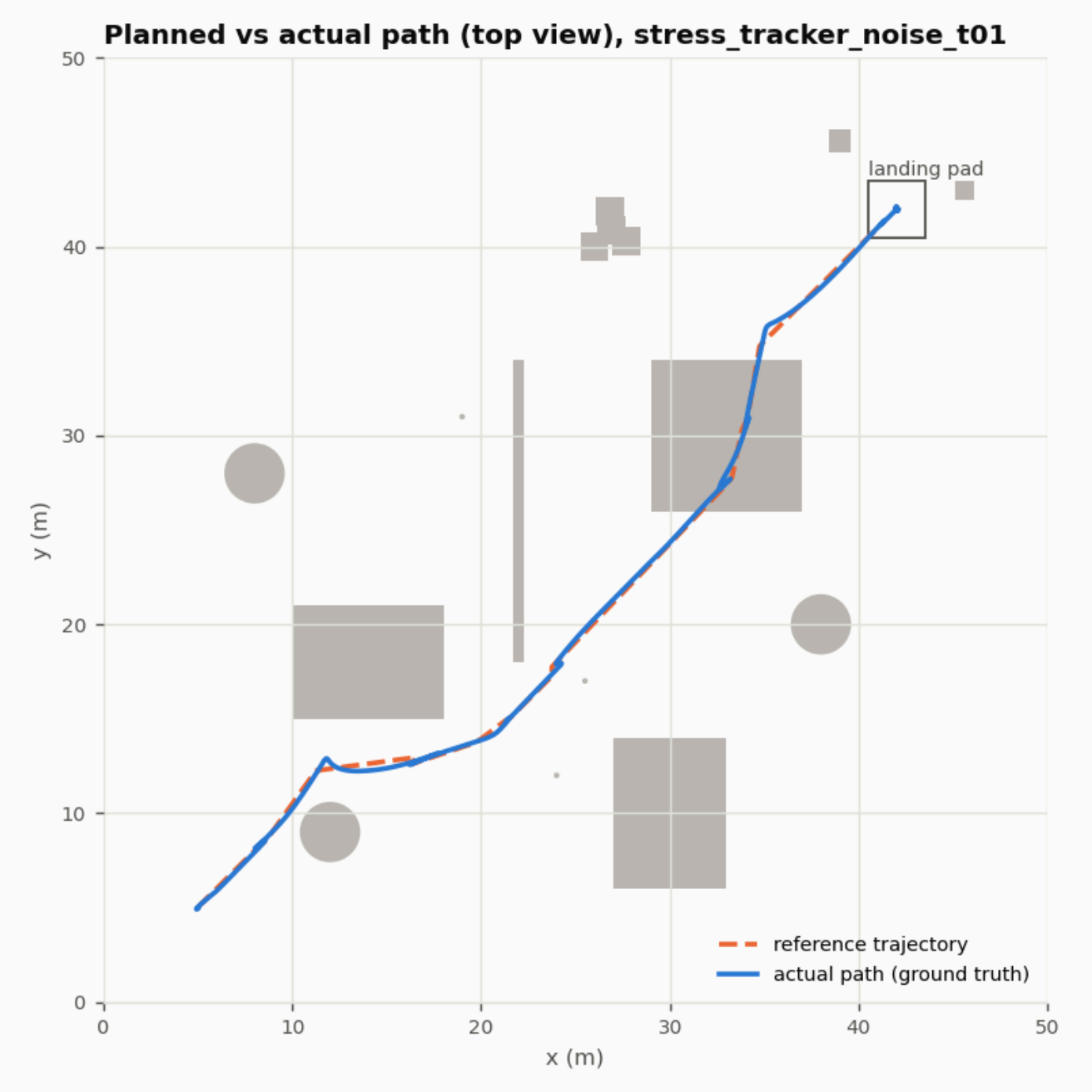

Planned vs actual path (top view), eval_baseline_t20
landing pad
reference trajectory
actual path (ground truth)
x (m)
y (m)
0
10
20
30
40
50


Planned vs actual path (top view), eval_dynamic_t10
landing pad
reference trajectory
actual path (ground truth)
x (m)
y (m)
0
10
20
30
40
50


Planned vs actual path (top view), stress_narrow_pole_t01
landing pad
reference trajectory
actual path (ground truth)
x (m)
y (m)
0
10
20
30
40
50


Planned vs actual path (top view), stress_overhead_t01
landing pad
reference trajectory
actual path (ground truth)
x (m)
y (m)
0
10
20
30
40
50


Planned vs actual path (top view), stress_timestamp_offset_t01
landing pad
reference trajectory
actual path (ground truth)
x (m)
y (m)
0
10
20
30
40
50



32. Robustness tests

Scenario	What is injected	Expected	Met	Landing error (mean)	Notes
`altitude`	Different cruise/approach altitude (takeoff 4 m, approach 7 m).	LANDED	100 %	3.9 cm	LANDED×1
`camera_blur`	Gaussian blur (9x9) on both cameras.	LANDED	100 %	2.7 cm	LANDED×1
`camera_exposure`	Under-exposure (gain 0.45) on both cameras.	LANDED	100 %	6.6 cm	LANDED×1
`camera_failure`	Failure injection - forward camera blackout for 8 s during navigation (fusion stops; mapping degrades to raw LiDAR).	LANDED	100 %	2.4 cm	LANDED×1
`camera_noise`	Additive Gaussian pixel noise sigma = 18 on both cameras.	LANDED	100 %	4.3 cm	LANDED×1
`crossing_obstacle`	Obstacle moving across the corridor while the UAV approaches.	LANDED	100 %	2.2 cm	LANDED×1; replans 6.0, replan latency 949 ms

Scenario	What is injected	Expected	Met	Landing error (mean)	Notes
`dynamic_obstacle`	Unexpected obstacle appears across the path ahead during navigation (detection + replanning latency).	LANDED	100 %	1.1 cm	LANDED×1; replans 4.0, replan latency 1193 ms
`extrinsic_perturbation`	Camera extrinsic perturbed by 5 cm and 2 deg yaw / 1 deg pitch (fusion projection error experiment).	LANDED	100 %	5.8 cm	LANDED×1; extrinsic shift 17.8 px, edge chamfer 10.6 px
`landing_offset`	Approximate landing-zone estimate offset 2.5 m from the true pad (precision must come from the tag).	LANDED	100 %	4.6 cm	LANDED×1
`lidar_dropout`	30 % of LiDAR returns dropped.	LANDED	100 %	5.2 cm	LANDED×1
`lidar_failure`	Failure injection - main LiDAR blackout for 4 s during navigation (SENSOR_FAILURE -> hover -> recover).	LANDED	100 %	5.9 cm	LANDED×1
`lidar_noise`	LiDAR Gaussian range noise sigma = 5 cm.	LANDED	100 %	3.1 cm	LANDED×1
`narrow_pole`	Thin 0.2 m pole placed on the corridor centreline (narrow-obstacle LiDAR detection + clearance).	LANDED	100 %	4.4 cm	LANDED×1; replans 4.0, replan latency 916 ms
`no_path`	Failure injection - approved corridor too narrow to pass building/tree obstacles -> NO_SAFE_PATH safe state.	ABORTED	100 %	NOT MEASURED	NO_SAFE_PATH×1
`overhead`	Overhead canopy forced onto the route (mission corridor restricted so the canopy region must be crossed in 3D).	LANDED	100 %	5.0 cm	LANDED×1
`pad_near_obstacle`	Permanent 4 m pole 1.2 m from the pad centre inside the clearance radius -> landing must be refused (safe abort).	ABORTED	100 %	NOT MEASURED	UNSAFE_LANDING_ZONE×2
`pad_obstruction`	Crate on the landing pad when alignment starts; removed after 15 s (unsafe pad -> abort -> climb -> search -> land).	LANDED	100 %	4.1 cm	LANDED×2
`start_position`	Different start position (12, 3).	LANDED	67 %	2.6 cm	LANDED×2, NO_SAFE_PATH×1
`start_yaw`	Start yaw 120 deg.	LANDED	100 %	7.5 cm	LANDED×1
`static_wall`	Extra wall panel placed across the approved corridor before take-off (static obstacle).	LANDED	100 %	4.8 cm	LANDED×1; replans 5.0, replan latency 1251 ms
`tag_loss`	Down-camera occlusion for 4 s starting 2 s after descent begins (target loss -> hover -> reacquire).	LANDED	100 %	3.8 cm	LANDED×1; reacquisition 3.0 s
`timestamp_offset`	Forward camera timestamps shifted by +80 ms (beyond max sync delta 50 ms) -> fusion rejects pairs, mapping falls back to raw LiDAR.	LANDED	100 %	4.1 cm	LANDED×1; sync abs(Δt) 20.0 ms
`timestamp_offset_small`	Forward camera timestamps shifted by +30 ms (within tolerance).	LANDED	100 %	3.9 cm	LANDED×1; sync abs(Δt) 30.0 ms
`tracker_noise`	Degraded ground tracking (sigma_r 0.15 m, sigma_angle 4 mrad, 10 % dropout) - localisation robustness.	LANDED	100 %	9.8 cm	LANDED×1

33. Failure modes

Failure modes found and fixed during bring-up (ground truth leaking into EKF2 through PX4's gz_bridge, uXRCE timesync vs. sim time, home latched before EKF convergence, gpu_lidar cube-map seam phantoms, landing-gear self-returns, tag too small / too large for the camera, A* corner cutting, inconsistent validation sampling, landing in place after an unsafe-pad abort) are listed with root causes in `docs/TROUBLESHOOTING.md`. Injected failures (LiDAR/camera blackout, timestamp offset, tag loss, unsafe pad, no path, extrinsic error, sensor noise) and their measured outcomes are in section 32.

Infrastructure: 1 run(s) were stopped by the simulator-stall watchdog (Gazebo sim time stopped advancing; not an autonomy outcome) and re-run once with the same seed: `stress_static_wall_t01`. Archived in `results/infra_failures/`.

34. Limitations

- Localisation uses a *simulated* external tracking station (as PS05 prescribes); its noise model is Gaussian and line-of-sight occlusion of the tracker is not modelled. On-board GPS-free odometry (VIO / LiDAR odometry) is not implemented.
- `start_position` met its expected outcome in 2 of 3 trials; failed run(s): `stress_start_position_t01` (outcome NO_SAFE_PATH, revision 1f9fe4e-dirty, 0 collisions). The failure was a safe abort (NO_SAFE_PATH: no route found from the stop point although one exists in the true geometry, i.e. the live map closed a narrow gap); the planner now dumps its exact inputs on every A* failure for diagnosis. Root cause not yet confirmed.
- All results are simulation-only (Gazebo Harmonic, lockstep PX4 SITL) on one world; no hardware or field data.
- Under load the simulator runs below real time and the forward camera renders below its 10 Hz target, so a fraction of LiDAR scans has no camera frame within the 50 ms sync window (those scans are mapped without colour). Python nodes bound the achievable rates.
- The planner treats unknown space optimistically; newly observed obstacles cause stop-and-replan cycles (safe but slower than a continuously re-optimising local planner).
- Weighted A* (w = 1.3) trades bounded sub-optimality for speed.
- The landing tag pose z is biased by ~0.2–0.3 m (texture scale); only the horizontal tag position is used and height comes from the downward LiDAR.

35. Future work

On-board LiDAR-inertial odometry as a second, independent localisation source with tracker-loss handling; continuous local trajectory optimisation (e.g. MPC / kinodynamic search) instead of stop-and-replan; C++ ports of mapping and planning; nested AprilTags for continuous visual guidance to touchdown; wind and sensor-latency stress tests; hardware-in-the-loop and field trials with the PS05 ground tracker.

36. Reproducibility

Version-locked Docker image (`docs/ENVIRONMENT.md`); world and vehicle generated from versioned YAML; every run stores its scenario, seed, full configuration, config hash and git revision. Commands: `./scripts/run_tests.sh`, `./scripts/smoke_test.sh`, `./scripts/run_full_campaign.sh 20 10 1`, `python3 scripts/aggregate_results.py`, `python3 scripts/generate_report.py`.

37. Conclusion

The system demonstrates, in simulation and with logged evidence, GPS-independent 3D navigation inside an approved corridor, LiDAR–camera fusion and 3D mapping, replanning around unexpected obstacles, and LiDAR-validated precision landing, together with explicit safe behaviour for every injected failure. The measured results above — not design targets — are the basis for any claim made from this work.